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Development of a BCI-based Virtual Assistive Keypad using OpenBCI Equipment

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ABSTRACT

Brain-computer interfaces (BCIs) enable direct communication between the brain and external devices. They are also relevant and useful for applications such as assistive communication and neurorehabilitation for injured patients. This Bachelor Thesis describes the development and implementation of a real-time BCI system that controls an assistive virtual keypad using non-invasive electroencephalography signals. The system comprises the Ultracortex Mark IV headset and OpenBCI Cyton board equipment, including eight EEG channels placed according to the 10-20 international system (Fp1, Fp2, C3, C4, P7, P8, O1 and O2), along with dry electrodes as well.

The acquisition setup has been systematically characterized and optimized through multiple experimental phases by evaluating signal quality, impedance, testing accuracy, electrode preparation procedures, filtering configurations, and noise rejection. The Steady-State Visual Evoked Potential paradigm was selected for the BCI implementation due to its low channel count and minimal calibration requirements. The signal processing pipeline involves bandpass filtering, Power Line Interference suppression, Common Average Reference application, spectral feature extraction, and frequency-based classification using Canonical Correlation Analysis. The proposed architecture is intended to be functional, scalable, and supported by the current scientific literature. This architecture is intended to be extended to more complex interfaces in future studies.

RESUMEN

Las Interfaces Cerebro-Computador (ICC) permiten la comunicación directa entre el cerebro y dispositivos externos, con aplicaciones en comunicación asistida y neurorrehabilitación en pacientes lesionados. Este Trabajo de Fin de Grado describe el desarrollo e implementación de un sistema ICC en tiempo real que controla un teclado virtual asistido mediante señales de electroencefalografía no invasivas. El sistema integra el casco Ultracortex Mark IV y la placa OpenBCI Cyton, con ocho canales de EEG y electrodos secos colocados según el sistema internacional 10-20 (Fp1, Fp2, C3, C4, P7, P8, O1 y O2).

La configuración de adquisición se ha caracterizado y optimizado sistemáticamente en múltiples fases experimentales, evaluando la calidad de la señal, la impedancia, la preparación de los electrodos y el rechazo del ruido. Se seleccionó el paradigma del potencial evocado visual en estado estacionario por su reducido número de canales y mínimos requisitos de calibración. El procesamiento de la señal incluye filtrado paso banda, supresión de interferencia de la red eléctrica, referencia promedio común, extracción de características espectrales y clasificación en frecuencia mediante análisis de correlación canónica. La arquitectura propuesta pretende ser funcional, escalable y estar respaldada por la literatura científica actual, y ampliarse a interfaces más complejas en futuros estudios.

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1 INTRODUCTION

The human brain is one of the main organs which regulates plenty of body processes. It constitutes the Central Nervous System (CNS), and is able to monitor digestive and breathing activities, voluntary movement coordination and thinking conscious reasoning [1]. These actions imply neuronal electrical activity being elicited, due to information transference through synaptic excitations that occur between nerve cells (neurons) that build up the brain tissue.

A Brain-Computer Interface (BCI) is a system that establishes a direct communication pathway with the human brain (see **Figure 1**), translating its output into commands that are carried out by external devices to achieve a desired action [2–4]. Therefore, they address the dissociation that may happen between the CNS with the neuromuscular system, which is linked to communication disabilities or motor activities due to a certain injury or disease. Indeed, this scenario occurs for individuals who are severely affected by motor disabilities such as amyotrophic lateral sclerosis, brainstem stroke, spinal cord injury or locked-in syndrome [4]. Among these sorts of conditions, it may happen that despite maintaining cognitive functions intact, their conventional communication channels are partially or completely impaired or even lost.

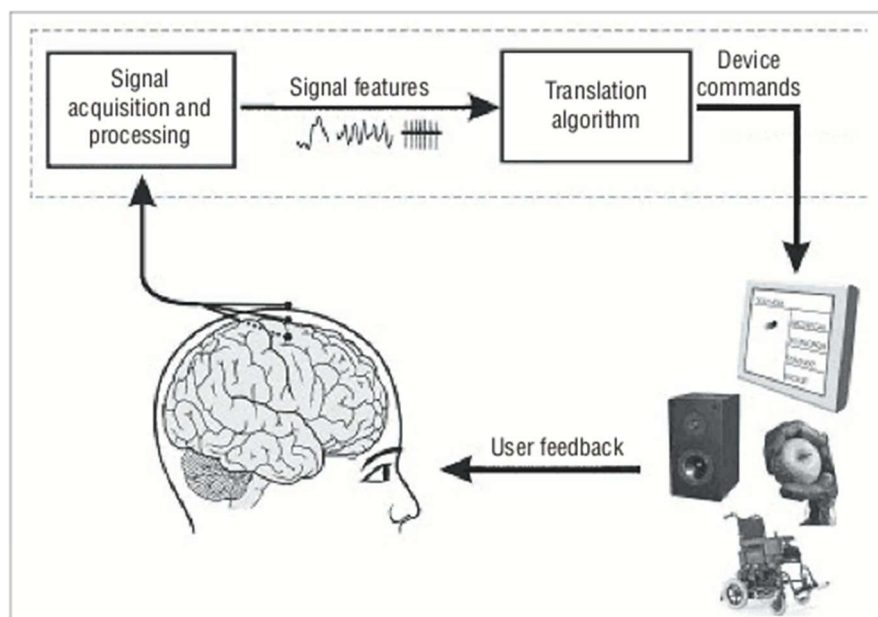


Figure 1 BCI model sketch [5].

The ability to communicate as well as interact with the environment is essential for human quality of life. Therefore, BCIs aim to restore, replace, enhance, supplement or improve the natural CNS brain signal activity in patients who suffer from the conditions mentioned above [2,3]. These systems are also able to establish a direct information transference pathway between the neurological activity and external devices, avoiding the entire neuromuscular system if it does not work properly and replacing it [4,6].

1.1 Brain Anatomy and Function

The brain is primarily made up of gray matter and white matter. The first one is located at its outer shell and comprises nerve cells, also named as neurons, which are responsible for performing brain signal processing. In contrast, the white matter gathers nerve axons, which are linked to the gray matter (brain cortex), the brain stem, and the peripheral motor and sensory units. Both elements mentioned comprise brain tissue and are interconnected by support connective tissue [1].

Nerve cells are distributed along the cerebrum, located in the upper part of the brain (see **Figure 2**). This structure processes the initial physical activity's learning pathway, directing the movement of the muscles. Furthermore, each cerebrum's section provides a different functionality: the frontal area manages speech, emotion, problem-solving, and skilled movements; the parietal lobe registers touch, pain and temperature sensations; the occipital section collects visual processes; and the temporal lobes are excited due to hearing scenarios or to store memory [1].

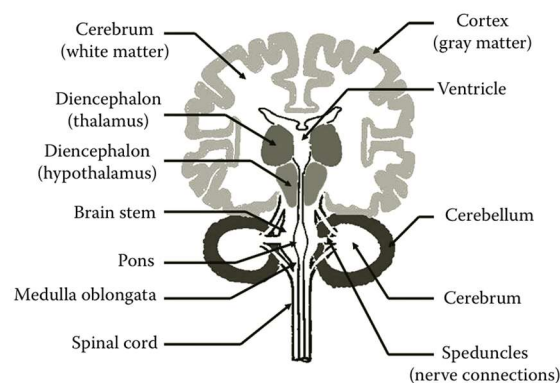


Figure 2 Schematic anatomy of the CNS [1].

Below the cerebrum is located the cerebellum (**Figure 2**), which is split into two halves such that the right hemisphere determines the left side of the body and the right side coordinates abilities regarding the left body area. Specifically, the left hemisphere oversees speech, local thinking or writing tasks [1], while the right half controls creativity achievements as well as artistic abilities. The cerebellum coordinates muscle motion as well as familiar actions linked to posture and balance maintenance.

1.2 BCI Classification Overview

BCI systems require neurophysiological devices to record brain signals. Hence, they can be categorized according to different criteria such as the recording method and its degree of invasiveness, the type of applied stimulation, or the brain signal pattern. Regarding the invasiveness of the recording method, non-invasive techniques are based on using superficial sensors like the ones placed on the skin or the scalp. This BCI type does not require any surgical or chemical implantation to record the neural activity. Some examples of non-invasive BCIs are electroencephalography (EEG), functional Magnetic Resonance Imaging, among others. On the other hand, invasive methodologies involve neuro-signaling techniques by which the electrodes make direct contact with the brain tissue, providing higher accuracy but increasing the risk for the safety of the individual. This application is achieved through recordings gathered such as from the electrocorticography technique [2].

BCIs can also be categorized as exogenous or endogenous. Exogenous BCIs require minimal training procedures, their setup tends to be simple, and they provide a high exchange of information bit rate. However, they may cause tiredness in users, and they imply permanent attention to the externally produced stimuli. Endogenous BCIs, on the other hand, operate more freely as they are independent from any stimulation and are more suitable for sensorimotor or cursor control applications. Despite these advantages, they offer a low bit rate, they are time-consuming as they require a lot of training procedures and they need many channels to achieve good performance [4].

Lastly, BCI systems can be synchronous or asynchronous. Synchronous applications are simpler in terms of performance and allow for avoiding or removing artifacts, but seem to follow an artificial interaction. On the other hand, asynchronous

BCIs do not require waiting for external cues or triggers, but, because of this, they implement more complex designs [4].

Regarding the type of BCI application, they may aim to develop communication or motion control. Regarding the first application, its purpose is generally to display a virtual keyboard on a screen for individuals with neurological or speech disabilities. The second type of application is oriented to patients who suffer from sensory or motor loss functionalities that generate dependency. Artificial muscle contractions can be elicited to solve this problem. An example would be to manage to steer a wheelchair for an individual with a spinal cord injury [4].

While BCI systems have experienced noticeable growth recently, their usage is limited by several barriers: the intra-subject and inter-subject variability of the generated EEG signals, the difficulty in achieving reliable signal quality, the computational complexity of real-time processing pipelines and the hardness of achieving implementations valid for accessible hardware [7,8]

1.3 EEG as the Acquisition Modality

Many non-invasive BCI types are based on EEG signals. These systems stand out due to their temporal resolution and low-cost conditions, as well as a complete absence of surgical risk as it employs non-invasive procedures [1,8]. Over the last few decades, EEG-based BCIs have evolved from laboratory trials to functional assistive devices (see **Figure 3**), with applications including augmentative communication, neurorehabilitation, environmental control and cognitive state monitoring [4,7,8].

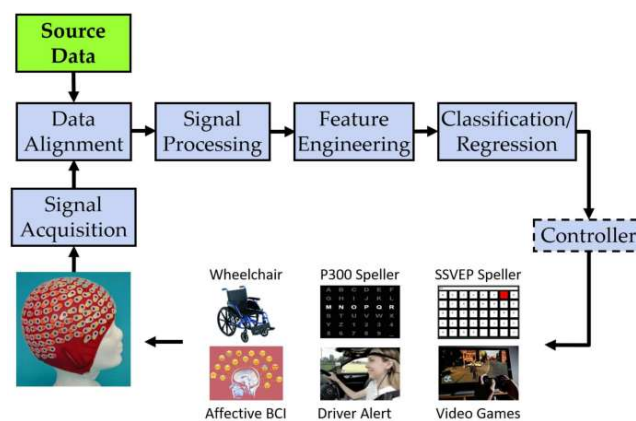


Figure 3 BCIs implementations [10].

EEG measures the summed post-synaptic action potentials generated by synchronized neurons fired during excitation, which are recorded as voltage differences between the scalp electrodes [9]. Because of the distance regarding the information flow through the brain tissue, the skull and the skin between the electrode's surface, the EEG cannot detect single neuron excitations [1]. Indeed, impedance signal values will even increase more due to the hair appearance at the scalp and the dead skin, which are bad conductors [1]. For these reasons, EEG recordings gather groups of neuron's current flows within the cerebral cortex area instead.

Neural oscillations are observed in EEG signals as brainwaves (see **Table 1** and **Figure 4**). Each one represents a specific level of brain activity and mental state. During EEG spectral analysis, the presence or absence of these waves is an indicator of a certain physiological condition to reach the appropriate diagnosis [1]. They are also relevant to characterize the amplitude of the signal to artifacts in a specific analyzed EEG task procedure. An example could be Berger's effect (increase of alpha wave amplitude) as a reference comparison between the Eyes Open and Closed conditions [10].

Table 1 EEG Frequency Bands characteristics description.

Activity Type	Frequency Band (Hz)	Description
Gamma	32 - 100	Linked to high-perception tasks (i.e. problem solving, alertness). Several areas of the brain are actively processing simultaneous information.
Beta	13 - 32	Indicates consciousness and active reasoning such as logical thinking. It is associated with a focused state regarding one task or a dynamic conversation.
Alpha	8 - 13	Associated with relaxation states. It is enhanced in creative achievements, meditation, and especially with the eyes closed.
Theta	4 - 8	Related to deep relaxation states like deep meditation or slight consciousness.
Delta	0.5 - 4	Implies late sleeping and slow scenarios.

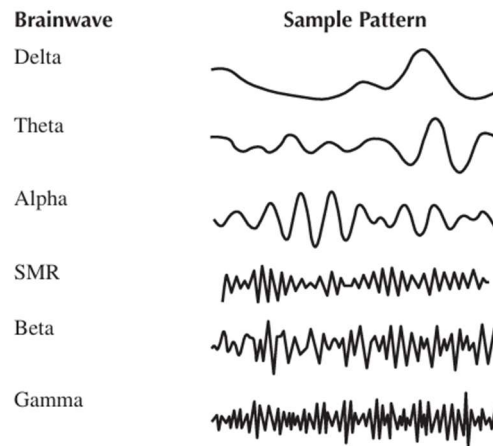


Figure 4 EEG brainwave types [2].

EEG recording procedures typically follow the electrode placement defined by the standard 10-20 International System (see **Figure 5**). This reproducible positioning method divides the skull into six areas from the nasion node (point between the forehead and the nose) to the inion node (skull bump) with interval rates of 10%, 20%, 20%, 20%, 20% and 10% of the complete distance between these nodes: the frontopolar (Fp), frontal (F), central (C), temporal (T), parietal (P) and occipital (O) [1,2]. Also, the Z (zero) refers to the electrodes placed on the midline. Furthermore, odd numbers are used for areas on the left hemisphere and even numbers for the right hemisphere, so that the smaller the number is, the closer it will be to midline Z. For any sort of EEG configuration, at least three electrodes are required: the ground, the reference and the recording electrode.

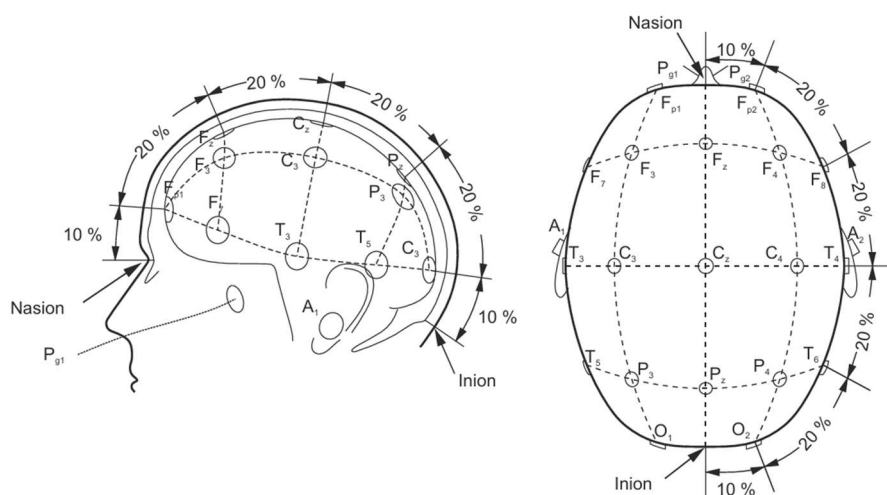


Figure 5 10-20 International System's EEG electrode placement [4].

EEG signals can be detected using dry or gel electrodes, which show significant differences in their contact impedances. As detailed by Sosa Rojo [11], dry electrode impedance values tend to rise up to 1000 k Ω , while gel electrodes may have impedance values close to 0 k Ω as they are used with a liquid that enhances the conductivity and lowers the impedance between the electrodes and the scalp. Furthermore, dry electrodes are associated with 84% reliability while gel ones are related to 97% [12]. This feature indicates that channel reliability is defined as poor if it exhibits strong artifacts among at least 50% of a specific chunk of data analyzed with respect to all the data sequences from a given channel. Another main difference is related to the electrode placement, as dry electrodes are easier and faster to place as they do not require any additional conductive gel for their functionality. However, this is at the expense of their sensitivity to artifacts, which is higher than that of gel electrodes [11].

1.4 BCI Communication Paradigms: Overview

The purpose of a BCI is to interpret human user intentions by monitoring their cerebral activity and cognitive tasks. Depending on the physiological phenomena response to a certain event, the results are classified as different BCI types that gather distinct characteristic signals [4]. The neurological responses elicited by visual, auditory or motor stimulations define different BCI paradigms, and tend to be used for non-invasive BCI systems with communication or control/motor application purposes [2]. The following sections describe BCI paradigms oriented to communicative applications, like the Visual Evoked Potentials or the P300 [2,4].

1.4.1 Visual Evoked Potentials (VEPs)

VEPs are brain signal modulations generated in the visual cortex after receiving a visual stimulus. They have a high information transfer rate (60-100 bits/min), enabling a wide selection of choices and do not require any training procedures [4]. These modulations are easy to detect when the stimulus involved is placed at the center of the user's visual field [4]. VEPs can be transient or steady-state:

1. Transient VEPs (TVEPs) occur when the applied visual stimulation frequency is below 6 Hz [4]. They are associated with abrupt flashing light changes like the onset/offset pattern, presenting as a result negative and positive peaks between

90-120 ms [4]. TVEPs are not used for BCIs as they are susceptible to variations due to electromyographic noise contamination.

2. Steady-State VEPs (SSVEPs) are elicited by the same stimulus, but with flickering frequencies higher than 6 Hz [4]. They may show a sinusoidal waveform with a fundamental frequency (see **Figure 6**) linked to the flickering stimulus that allows users to select targets by eye-gazing. Depending on the stimulus modulation, they can be classified as time-modulated VEP (t-VEP), frequency-modulated VEP (f-VEP) and pseudorandom code-modulated VEP (c-VEP) [4]. While t-VEP and c-VEP rely on time-differing or pseudorandom ON/OFF sequences, the f-VEP assigns each target a unique periodic frequency (fundamental and its harmonics), requires no training, enables a simple configuration and provides a high information transfer rate (30-60 bits/min), being suitable for few-option selection [4].

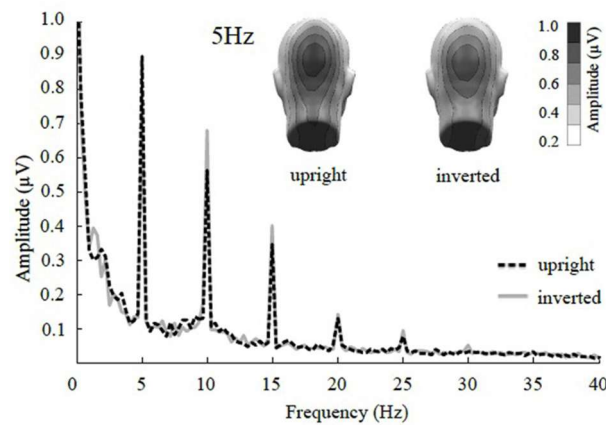


Figure 6 Steady-State Visual Evoked Potential [2].

The most typical VEP BCI application displays flashing stimuli (digits or letters) on a computer screen, inducing SSVEPs while the user gazes at the desired symbol. This control signal does not require training, but the user must fixate on one point. For this reason, SSVEPs are not suitable for advanced atrophies or degenerative stages of patients, as they cause eye-tiredness and are not aimed at scenarios involving involuntary movements [4].

The SSVEPs paradigm exploits the cortical responses elicited by periodic visual stimulation: when an individual gazes at a flickering frequency f (equal to 5 Hz in

Figure 6), the visual cortex generates an oscillatory response at an f frequency and its harmonics ($2f$, $3f$, ...), which is detectable in the occipital and parietal EEG channels [13,14]. A unique stimulation frequency is assigned to each target, so that the system selects it when its associated flickering frequency dominates the user's EEG spectrum. SSVEPs are characterized by acquiring a high signal-to-noise ratio (SNR) when targets are appropriately chosen and their performance is increased when combined with classification algorithms [13–15].

1.4.2 P300

This BCI type is linked to Event-Related Potential (ERP) phenomena such that positive voltage deflections are elicited around 300 ms after an infrequent stimulus appearance within a sequence of repetitive non-target stimuli (oddball paradigm), as shown in **Figure 7** [11]. These positive peaks can be evoked by infrequent auditory, visual or somatosensory stimuli [4]. The information transfer rate of these BCI systems is medium (20-25 bits/min), as it does not require training procedures, and it enables a high number of choices [4]. Despite not requiring training procedures, its performance may increase if it is combined with previous training [2,16].

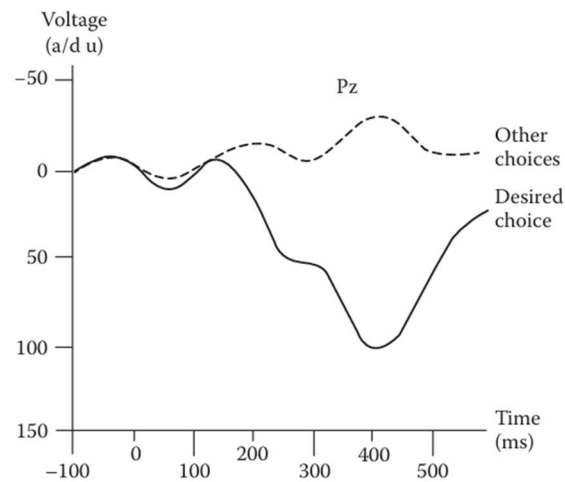


Figure 7 Typical waveform of a P300 ERP [2].

The most typical application for this BCI paradigm consists of a matrix of letters, numbers or any kind of commands that are flashed and gazed at by the user. The detection of a single trial comprises a low signal-to-noise ratio EEG signal [4]. For this reason, the rows and columns cells must be flashed several times for each choice, decreasing the

number of choices per minute (low information transfer rate and accuracy). However, performance can be enhanced with matrix visualization optimization or a more complex classifier implementation. As a reference, the first P300 matrix speller BCI was introduced by Farwell and Donchin [17], and like other subsequent studies, they considered this algorithm as valid and with a well-established signal processing pipeline.

1.5 Motivation and Objectives

1.5.1 Motivation

The main motivation of this Bachelor Thesis is to increase the quality of life of those patients who have lost communication abilities or gained motor disabilities due to injuries or disease and, thus, need an external tool to interact with the environment. Among the possible EEG-based BCI systems that can be developed to fulfil this purpose, a system with a virtual assistive keypad with symbols of different basic needs that registers the user's intention through neurological signals could be highly valuable. Particularly, the system could enable patients to indicate family members or clinical staff whether they are hungry or cold, to call emergency services due to pain or any concerning scenario and to request for help to go to the bathroom. The system would not require any physical interaction with the user interface to select the cell targets, as this would be accomplished just by gazing at the screen monitor while utilizing the headset.

The virtual assistive keypad described can be implemented through different BCI paradigms. Among them, SSVEPs offer several advantages compared to other paradigms such as the P300, as it does not require training, provides a high signal-to-noise ratio and information transfer rate per iteration, and relies on the occipital and parietal channels available in the hardware used. These features make it a suitable option for a low-cost system developed under limited constraints.

1.5.2 Objectives

The main objective of this thesis was to develop and validate a functional, low-cost, dry-electrode, real-time assistive BCI keypad based on the utilization of the open-source reference for BCI academic research: the OpenBCI equipment. This general objective is split into the following specific objectives:

1. To define the optimal experimental conditions for acquiring EEG signals using OpenBCI equipment, and to analyze the performance of dry electrode use, especially with respect to preparation time, conductivity and displacement.
2. To assess the EEG recording system's sensitivity to detect changes in neural activity against external stimulation.
3. To design and optimize a virtual assistive keypad based on the SSVEP paradigm through offline tests and analyses.
4. To implement and validate the BCI system in real time, providing visual feedback for the four assistive actions and evaluating its performance under the hardware, timing and computational limitations of the setup.

2 MATERIALS AND METHODS

The assistive communication SSVEP BCI system developed integrates the OpenBCI hardware equipment and open-source signal processing software. This hardware is described along section 2.1 and specifically comprises the EEG headset (see section 2.1.1), the OpenBCI acquisition board (see section 2.1.2) and the USB Dongle for laptop connection (see section 2.1.3). With respect to the software application (see section 2.5), a combination of Python signal processing modules gathered at the backend (see section 2.5.2) was included as well as a browser User Interface with HTML, CSS and JS modules gathered at the frontend (explained in section 2.5.3). Both parts of the project were interconnected via WebSocket use (see section 2.5.3).

2.1 Hardware and Software Acquisition

2.1.1 EEG headset: Ultracortex Mark IV

The *Ultracortex Mark IV* EEG headset from OpenBCI (see **Figure 8**), built in previous projects through 3D printing procedures, was used to conduct experiments. It contains dry electrodes of three types following the 10-20 standard configuration: the Spikey Units, for high hair density areas (C3, C4, P7, P8, O1 and O2); the Comfort Units, responsible for distributing the helmet's weight (Fz, Cz and Pz) and the Flat Units, placed at non-hairy areas nodes (Fp1 and Fp2).



Figure 8 Ultracortex Mark IV placement (left) and display (right).

For the REFERENCE and GROUND electrodes, ear clips were used, connected to each earlobe respectively (see **Figure 8**, left). These ones correspond to the reference node on the left side and to the bias node serving as the ground with common-mode noise rejection on the right side. These electrodes were implemented instead of gel electrodes due to conformity and utility reasons, as they require lower preparation time.

2.1.2 Acquisition board: OpenBCI Cyton

An OpenBCI Cyton board was used for signal acquisition, which allowed recording 8 EEG channels and is regarded as an affordable biosignal acquisition system [18]. It contains an ADS1299 analog frontend chip, designed by Texas Instruments [19], which enables the analog-to-digital conversion (ADC) of data. This chip has a high signal-to-noise ratio (SNR) of around 121 dB, according to the specifications from the official OpenBCI Documentation [20], and a voltage resolution of 0.298 $\mu\text{V/bit}$ or 0.022 $\mu\text{V/count}$. This ADC chip was integrated with a PGA (Programmable Gain Amplifier) of 24 bits [19].

Communication between the OpenBCI Cyton board and the computer was established through an FTDI-based USB dongle, which provides a USB-to-UART interface for continuous serial data streaming. The exchange of EEG data was processed by the project via BrainFlow, which is a native set of Python libraries compatible with the Cyton board.

The leads of each electrode were connected to the Cyton board pins shown in **Table 2**. Then, the REFERENCE electrode (left ear clip) was connected to the SRB2 pin from the Cyton board. It was also attached to the earlobe, as adipose tissue is not electrically conductive. The ground electrode (right ear clip) was attached to the BIAS pin, corresponding to noise cancellation. While placing the Ultracortex Mark IV, it was ensured that it was oriented in the appropriate position: the Spikey Units and the Cyton board were placed at the nape (inion node), and the frontal electrodes which are Comfort Units, faced the same direction as the nose (nasion node).

The following **Table 2** and **Figure 9** indicate the Cyton board PIN connections with the EEG arrangement used. The 1-8 channel electrodes considered were attached to

the pins ranging from N1P-N8P (**Figure 9**, left). The white cable was linked to the bottom SRB Pin (**Figure 9**, right), connected as well to the left ear clip REFERENCE electrode. The black wire (**Figure 9**, right) was attached to the BIAS Cyton board pin, which was also connected to the noise-cancelling or GROUND node right ear clip.

Table 2 Electrodes used node connections to the associated Cyton Pins.

Electrode Node	Cyton Pin
Fp1	N1P
Fp2	N2P
C3	N3P
C4	N4P
P7	N5P
P8	N6P
O1	N7P
O2	N8P
REFERENCE	SRB
GROUND	BIAS

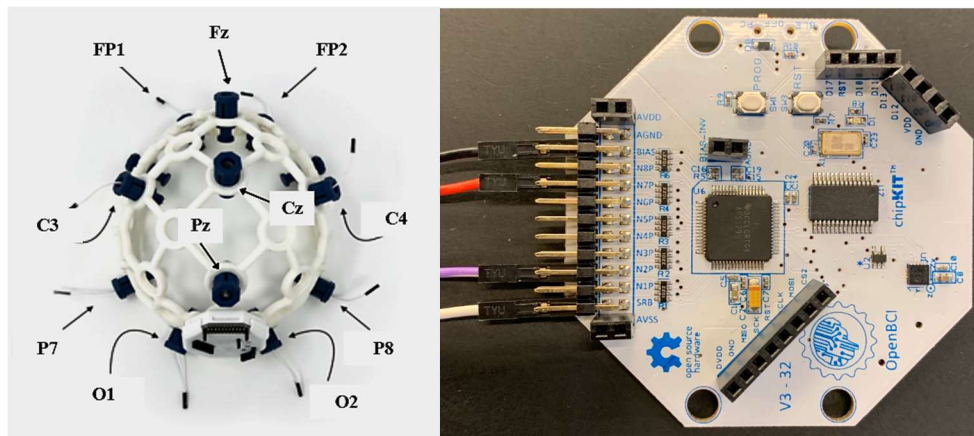


Figure 9 Electrode's locations (left) and reference and bias connections to the Cyton board (right).

The acquisition equipment used also included a Daisy board, which enabled recording of 8 additional EEG channels. However, it was not used as the signal quality as some of its channels were poor. Therefore, a plastic isolating bag was used to cover the Daisy board and avoid interference, as shown in **Figure 10**.

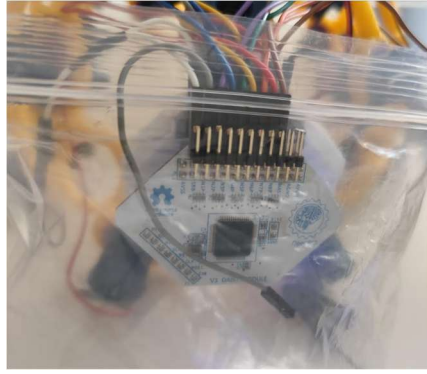


Figure 10 Isolated OpenBCI Daisy board with a plastic bag.

2.1.3 USB Dongle

The USB Dongle was an intermediary between the Cyton board and the used laptop through the linked port between them. When it is connected to the PC, an LED on the Cyton board emits a blue light (see **Figure 11**). Also, when data streaming occurs, another LED on the board emits a red light.

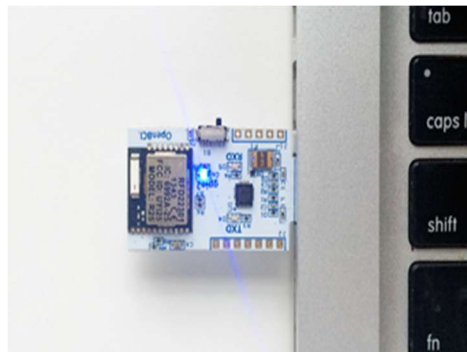


Figure 11 OpenBCI USB Dongle connected to the laptop.

The USB Dongle also contains a slide switch with two possible positions: GPIO6 and RESET. When the switch is moved to the GPIO6 position, data is sent from the Cyton board. Otherwise, when the switch was set to the RESET position, the Dongle was reprogrammed. When the USB dongle did not transmit data, it was disconnected, as well as the Cyton board, and plugged in again to register EEG data correctly.

2.1.4 OpenBCI Graphical User Interface (GUI)

Once the Cyton board was configured, the headset adjusted and the USB dongle attached to the PC, the OpenBCI Graphical User Interface (GUI) exe file was launched.

The first screen of the OpenBCI GUI showed different options to perform the main recording configuration steps. In this screen, the LIVE (from Cyton) button was selected from the DATA SOURCE section, and the PICK TRANSFER PROTOCOL area, specifically the Serial (from Dongle), was chosen (see **Figure 12**).

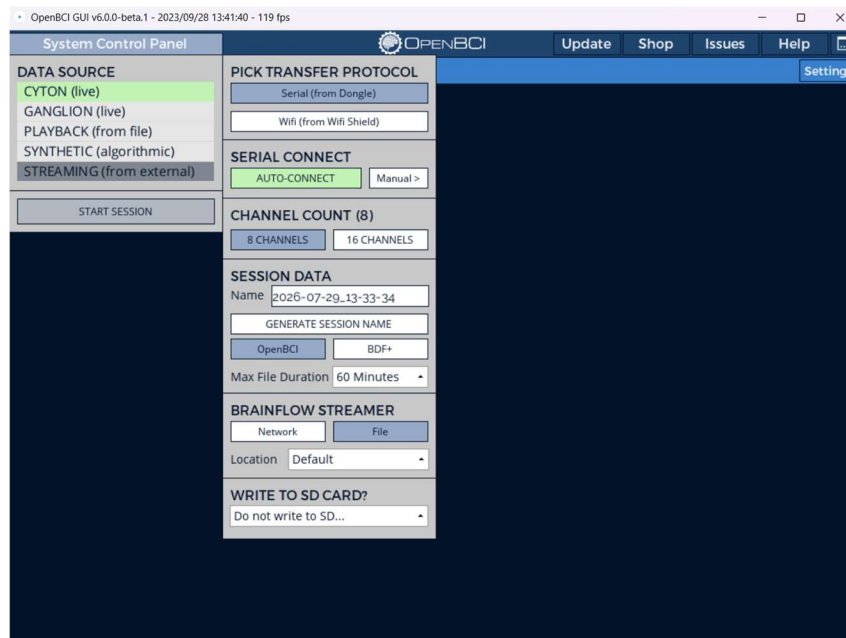


Figure 12 OpenBCI GUI System Control Panel configuration.

To automatically detect and configure the serial port associated with the Cyton board, the Manual button from SERIAL CONNECT was selected (**Figure 12**). Since a Windows operating System was used for this Bachelor's Thesis, the port type was COM*; otherwise, with Mac or Linux devices, the serial port appeared as /dev/tty*. When troubleshooting occurred while attempting to find the OpenBCI Cyton port, the AUTO-SCAN option was used, which displayed the available ports if the Dongle was switched to the GPIO6 pin. Finally, the CHANNEL COUNT was set to 8, which was the number of EEG channels recorded. Once the most crucial configuration steps were performed, the START SESSION button was selected.

When the START SESSION button was clicked, the GUI showed different windows with information from different visualization widgets, which were the Time Series, FFT plot and accelerometer by default. The Time Series widget represented signals in the time domain (**Figure 13**), which allowed checking their amplitude range and whether a certain channel was railed; and the FFT plot showed the Power Spectral

Density (PSD) of the recorded signals (**Figure 13**), which was analyzed to determine the Power Line Interference (PLI) appearance or removal. Another widget included in the GUI, but not shown by default, was the Cyton Signal (**Figure 14**), which was used to check the electrode-to-skin impedance of each channel before starting the recording session (see section 2.1.5). After using it, the protocol suggested pressing the Start Data Stream to register EEG signal recordings (**Figure 14**).



Figure 13 Data Stream recording visualization.



Figure 14 Initial launched GUI session.

The GUI also included a filter configuration option, as shown in **Figure 15**, which enabled the selection of the digital filtering technique for conditioning the EEG signal input. It was used to adjust the Bandpass filter specifications (order, type and frequency limits) and the Notch filter, which was applied to reduce the PLI noise (50 Hz in the European Union).

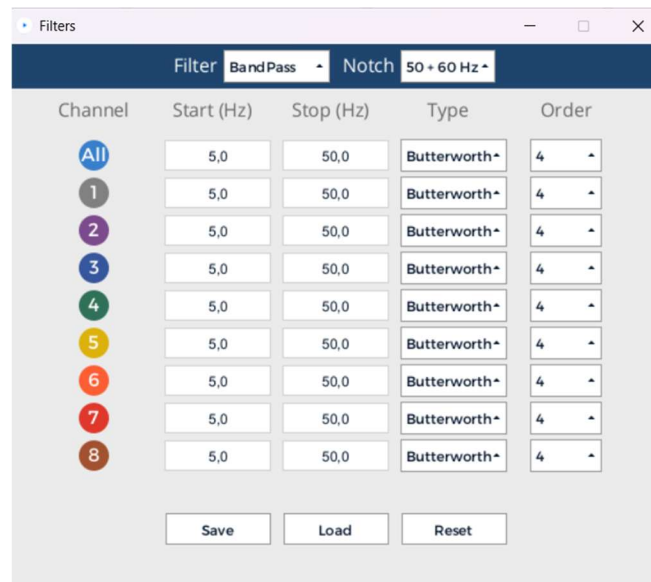


Figure 15 Default GUI filter configuration.

Data from each recording session was saved as a text file in CSV format, whose name contained the timestamp automatically generated at the end of the streaming session, indicating both the day and time of the recording session.

2.1.5. Electrode-to-skin-impedance check

Electrode-to-skin impedance was measured before data stream acquisition using the impedance testing GUI widget and the default bandpass filter of the program (5-50 Hz). When this tool was used, a controlled injected current (31.2 Hz, 6nA) was elicited between each recording electrode and the reference electrode [21]. In **Figure 16**, a significant peak around 30 Hz can be observed corresponding to the impedance measured at channel 2 due to its purple appearance. The voltage difference between the two nodes was measured and used to calculate the impedance value of the recording channel using Ohm's law. Data streaming could not be started while impedances were tested.

The measured impedance values were displayed by the OpenBCI GUI in three different colors depending on their value. Default impedance thresholds between colors were used, so that impedance values below 750 K Ω were shown in green, those in between 750 K Ω and 2500 K Ω in yellow and those above 2500 K Ω in red.



Figure 16 Impedance Testing visualization.

2.2 Signal acquisition

Several tests were performed with acquisition conditions related to the table in the first Results section (see **section 3.1 Identification of optimal acquisition conditions**). After identifying the optimal conditions until low impedances and higher signal stability were gathered, the results described in the section mentioned were obtained. These conditions were contemplated for the rest of the study detailed in the following sections.

Regarding the signal acquisition configuration, the sampling frequency of the OpenBCI Cyton was 250 Hz (250 samples per second). Eight acquisition initial channels were utilized (Fp1, Fp2, C3, C4, P7, P8, O1 and O2). Bandpass Butterworth filters of 4th order ranging from 0.5-45 Hz were set at the GUI. This configuration was also considered

for Time Series visualization, FFT plot and impedance checking procedures. A Notch digital filter at 50 Hz was also included at the GUI to soften or even remove the PLI.

The values stored in the .txt files were already expressed in microvolts (μV), as converted by the OpenBCI GUI. Signal acquisition from the Cyton board was performed through the OpenBCI GUI, which relies on the BrainFlow libraries to communicate with the Cyton board and streams the acquired raw data. Each recording session was stored by the GUI as a .txt file, which was later loaded into Python scripts for further analysis.

The bandpass and the notch filter mentioned were applied only to the signals displayed on the GUI widgets (Time Series visualization, FFT plot and impedance check). Since the Cyton board does not include any analog filter, the signals stored in the recorded .txt files were saved as raw signals, without any digital filtering application. For this reason, since the recorded data was raw, the digital filtering required for its analysis was applied afterwards through the Python Jupyter Notebook scripts as described in the corresponding signal processing section (see 2.4.3).

2.3 Preliminary assessment of EEG responsiveness

The EEG registered signals were first evaluated under determined conditions to verify that the signal that was visualized was coherent with certain OpenBCI GUI widgets (see section 2.3.1) and through the eyes protocol under several conditions (detailed in sections 2.3.2 and 2.3.3).

2.3.1 Stimulation feedback

Before the eyes condition protocol, the evaluation of the EEG quality due to external stimulation was first verified utilizing two OpenBCI GUI widgets: the Head Plot and the Focus Widget. The Head Plot displayed the potential fluctuations at each channel from an upper-scalp view and was used to check that the channels reacted to visual and auditory stimulation. The Focus Widget was configured in relaxation or focus mode and provided auditory feedback to confirm that neurological responses were elicited, particularly with respect to the frontal area. These initial checks ensured that the headset and user responded coherently to stimulation before proceeding with the recordings described in the next section.

2.3.2 Eye condition protocol

Experiments were conducted based on the alternation between Eyes Open (EO) and Eyes Closed (EC) conditions for 11 minutes (see **Figure 17**). A timer was used to indicate that each minute had passed, so that the user changed from EC to EO and vice versa. Five epochs for the EC condition and five for the EO condition were considered, apart from the initial minute used as an electrode-scalp stability margin (see section 3.1 from Results for further understanding). During the EC scenario, the user remained relaxed, sitting on a chair, waiting for the beeping timer signal to open their eyes. Next, the user stared at a white wall until a minute was completed, alternating to the next epoch. This process was repeated until the 11 minutes were reached.

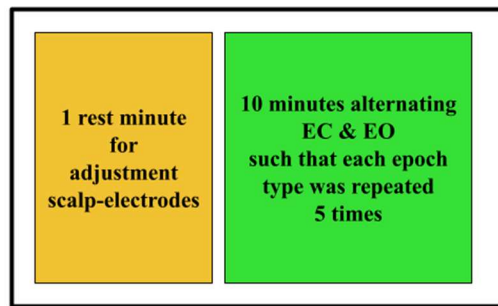


Figure 17 Trial procedure sketch.

2.3.3 EC/EO Signal Processing

The raw data automatically saved by the OpenBCI GUI for each recording session was uploaded into Python Jupyter Notebooks for further analysis. The signals obtained were bandpass filtered with a 4th-order Butterworth filter (0.5-45 Hz) to determine the ROI (Region of Interest). Since its upper cutoff (45 Hz) lies below the PLI, the 50 Hz interference was already removed by this filter, so no additional notch was required.

PSDs were estimated through Welch's method (windows of 5 s) such that 1250 samples were gathered with 50% overlapping in order to yield a frequency resolution of 0.2 Hz. For each channel, the signal was divided into overlapping segments weighted by a Hann window. This window type consists of a smooth bell-shaped function that avoids distortions resulting from the introduction of abrupt cuts during the recording sessions,

especially between EO/EC transitions. Periodograms were averaged to obtain the final PSD normalized to its maximum value.

Relative energy was computed for the five frequency bands: Delta (0.5-4 Hz), Theta (4-8 Hz), Alpha (8-13 Hz), Beta (13-32 Hz) and Gamma (32-45 Hz). The obtention of this representation was calculated as the area under the normalized PSD within the corresponding frequency range. This energy was conducted across all EEG channels, comparing also the Eyes Open and Eyes Closed conditions.

Spectrograms were also computed to express the variations of the spectral energy over time. Short-Time Fourier Transform was applied over consecutive Hann windows and 50% overlapping. A shorter window (~ 2 s, 512 samples) was chosen to preserve temporal resolution while keeping 0.5 Hz of frequency resolution. The resulting time-frequency representations were expressed in dB to display the energy differences in neural activity between the EO and EC conditions over time.

For the subsequent SSVEP system (described in section 2.4), the PSD analysis window was later reduced to 4 s, in accordance with the literature [13].

2.4 BCI design and offline assessment

Several experimental attempts were made until an affordable and functional system was reached, described along this section 2.4. Initially, the developed system just gathered the acquisition of the entry EEG signals. A .txt file through a recording module registered this raw data, which was uploaded next to Jupyter Notebook scripts where it was analyzed, preprocessed and classified. When the system's performance was validated and optimal it was switched to the online implementation (see section 2.5).

To develop the BCI design, the framework from Schalk et al. [22] named as BCI2000 was considered. It constituted a methodological reference for the modular signal processing pipeline: acquisition step, signal processing procedure, feature extraction application and signal classification as depicted in sections 2.4.2-2.4.5.

2.4.1 User Interface

The selected visual stimulation applied to the User Interface (UI) oscillated within the 8-15 Hz range, linked with alpha activity and excluding muscular interference associated with further frequencies. It was also related to better SSVEP response [15]. Four distinct cells were displayed flickering between black and white colors, gathering the following frequencies: target 1 corresponded to 8.57 Hz, target 2 to 10 Hz, target 3 to 12 Hz and target 4 to 15 Hz; resulting from the division between the 60 Hz screen refresh rate with integers (60/7, 60/6, 60/5 and 60/4) [23].

The visualization of each target corresponded to the indication of a certain desired assistive action or need for the patient. Specifically, the first target was represented as the desire to get warmer, the second as the need to get colder due to being too warm, the third cell represented the action of being in a lot of pain and the last one of requesting to go to bed due to being tired. Each frequency value produced a distinct response, which was gathered by the classifier algorithm (see section 2.4.5). This onset/offset pattern was synchronized with the refresh rate of the laptop used (60 Hz) [15,24].

Two interface layout design approaches were contemplated: displacing four target cells in a row-shaped layout as seen in **Figure 18**, or in a cross as depicted in **Figure 19**.

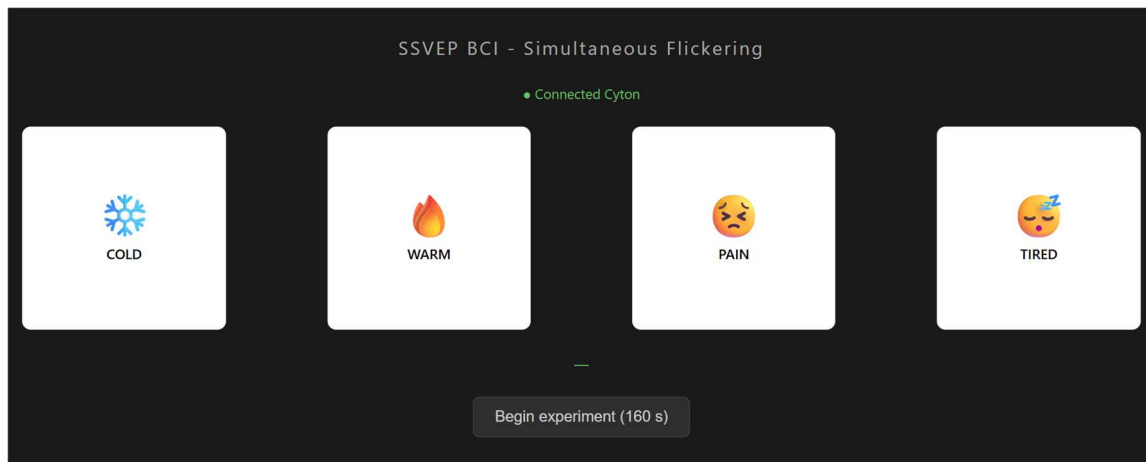


Figure 18 Row-shaped arranged cells layout.

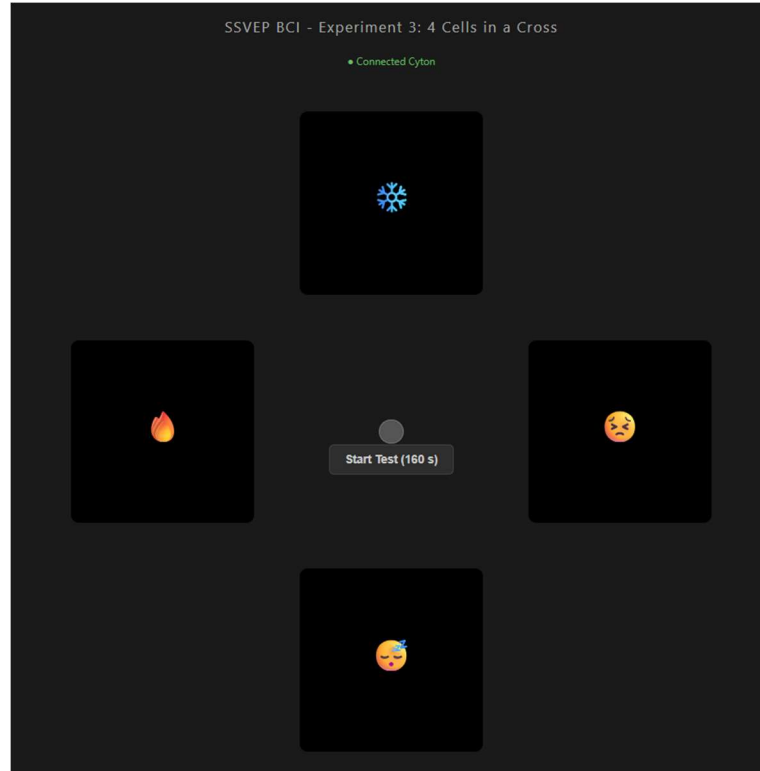


Figure 19 UI cross-shaped visualization.

2.4.2 Signal acquisition protocol

The first offline trial implementation was based on arranging four cells in the same row where the user gazed at each target sequentially for 160 s in total from left to right (see **Figures 20** and **21**). It was tested the user's neurological response to visual stimulation by maintaining all cells flickering simultaneously. An additional timer was used to indicate to the user that at each beep it must look at the next consecutive cell as well as a message to confirm which cell was intended to be gazed at. The total duration of the user's fixation time to the screen monitor for each cell was 40 s (see **Figure 20**). As observed in **Figure 21**, 5 s remained looking at the Pain cell before gazing at the Tired cell.

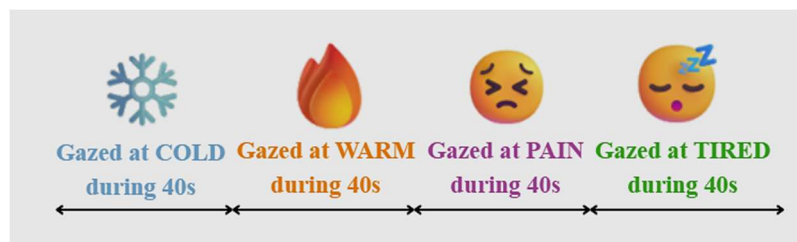


Figure 20 Recording offline protocol scheme.

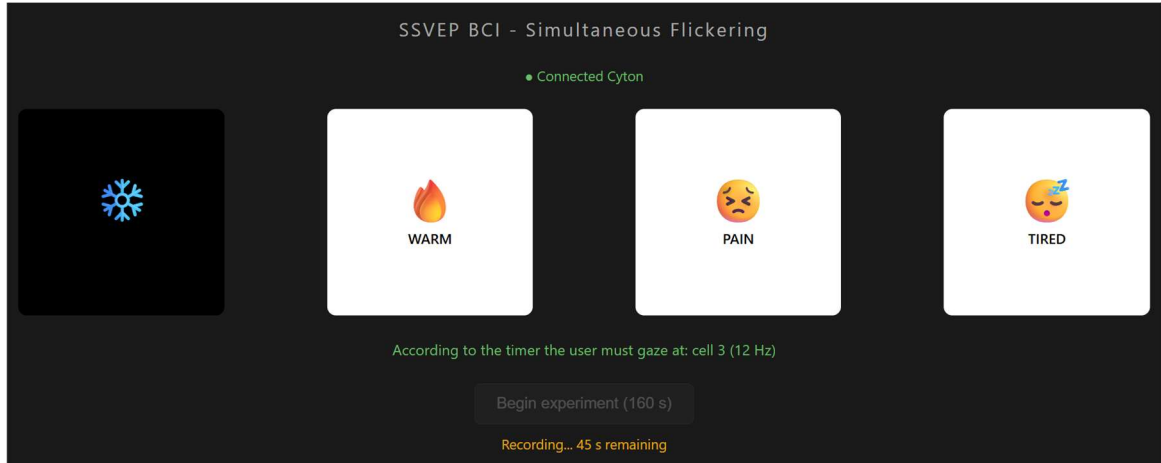


Figure 21 Row cell recording session visualization.

The other offline implementation considered was the one related to the cross-shape cell displacement (see **Figure 22**) as recommended by scientific literature [25]. Recording sessions were based on the user looking at each cell for 40 s from up to left, then to right and finally downwards during 160 s in total.

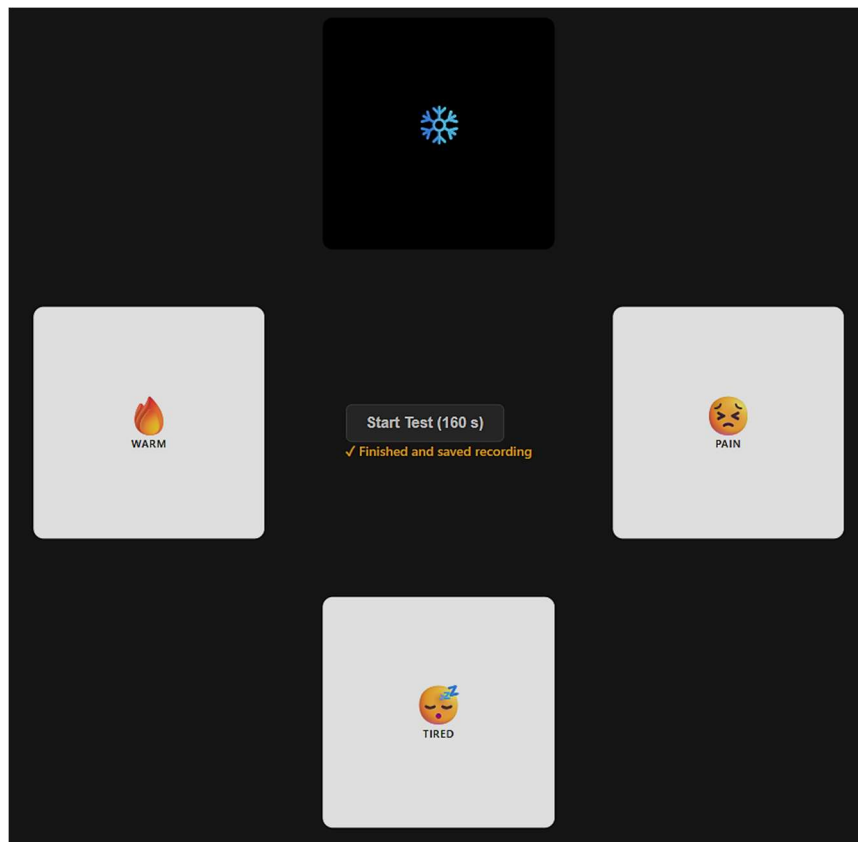


Figure 22 End of recording session.

2.4.3 Signal preprocessing

This step was used for cleaning the entry raw EEG signal from artifacts to analyze it and enable processing and classification next (see sections **2.4.4** and **2.4.5**). A bandpass filter between 7-70 Hz was implemented to gather the SSVEP-used frequencies (8.57-15 Hz), including its harmonics (up to 60 Hz for the 15 Hz target). The lower cutoff at 7 Hz, just below the lowest stimulation frequency (8.57 Hz), removed low-frequency drift and ocular artifacts, while the upper cutoff at 70 Hz excluded higher-frequency muscular (EMG) noise. The window size was about 4 s according to literature [13]. A comb filter was also applied to the signals to remove the PLI (50 Hz) and its harmonics (100 and 150 Hz). Eight acquisition initial channels were considered, but finally four were processed for user interface target selection (P7, P8, O1 and O2). CAR (Common Average Reference) was implemented, based on averaging the four EEG entry channels and subtracting from the recording electrode, identifying the useful signal in noisy recordings. By only analyzing four channels (parietal and occipital ones), the electrooculographic (EOG) artifacts from the frontal electrodes were excluded.

Through Python scripts, double digital filtering was applied to avoid interference and to clean the raw EEG entry data. This was included as the Cyton board does not contain analog filters and was only applied digitally through the OpenBCI GUI and for the assistive SSVEP BCI system were also applied. Once the registered raw data was preprocessed, it was represented through the following graphs: temporal visualizations, PSDs (to ensure that the target f_0 and its three following harmonics were registered as a visual stimulation response), analysis of summed PSD classifier (see section **2.4.4**) and CCA classifier (section **2.4.5**).

2.4.4 Feature Extraction: Summed PSD

The first feature extraction method was based on the power spectral density of the acquired EEG signal. For each analysis window, the PSD was estimated through Welch's method using 4 s Hann windows with 50 % overlapping, yielding a frequency resolution of 0.25 Hz through recordings.

For each candidate stimulation frequency f_0 (8.57, 10, 12 and 15 Hz), the spectral energy was summed over a band of 0.5 Hz centered on the fundamental frequency and its

two following harmonics (2f and 3f), obtaining one summed PSD value per candidate frequency. The candidate frequency with the highest summed PSD was intended to be identified as the user's target, as a stronger SSVEP response concentrates the signal energy at the stimulation frequency and its harmonics. Total energy visualizations were aimed at parietals and occipital channels and considering neurological frequency bands.

2.4.5 Feature extraction: Canonical Correlation Analysis (CCA)

For this second feature extraction it was considered the Canonical Correlation Analysis (CCA) algorithm. This one also enabled target classification. It is one of the most adopted classification methods for SSVEP BCIs [13]. It is a statistical method that calculates correlations between variables, finding the linear combinations of different multi-dimensional datasets and their weight vectors. Considering this information, it gathered the maximal mutual data correlation.

This method represented the X matrix as the raw EEG signals from all channels and the Y matrix as their reference signals (set of sinusoidal components and harmonics with respect to the fundamental frequency) [14,15,26]:

$$Y = [\sin(2\pi ft), \cos(2\pi ft), \sin(4\pi ft), \cos(4\pi ft), \sin(6\pi ft), \cos(6\pi ft)]$$

CCA was analyzed by assigning reference signals resulting from the combination of sinusoidal functions with their fundamental frequency and harmonics to EEG entry signals. The first canonical correlation component was applied to each EEG window. The estimated signal was compared with the values linked to the user's visual stimulation frequency response.

The CCA discrimination was graphically represented with 4 s windows through box plots indicating the average and standard deviation with respect to the canonical correlation coefficient and the first three harmonics, representing also outliers and false positives for each candidate's frequency.

2.5 Real-time BCI Implementation

Once the optimal parameters were identified in the offline assessment regarding the row-selected layout and the signal processing pipeline detailed in the previous section 2.4, the system was implemented in real time. This last stage of the project consisted of developing an online BCI system in which each cell represented a vital assistive function. These targets were designed to support both patients and their care providers, whether related to motor or speech impairments.

2.5.1 Recording Protocol

At the web User Interface, the left cell screen option indicates that the user requests eating or drinking, symbolized by a plate with cutlery as seen in **Figure 23**. The next symbol represents the body temperature according to the season; during summer may be linked to warm temperatures with a flame icon and during winter, to cold with a snowflake. As this project was delivered close to the autumn season, the snowflake symbol was implemented on the screen. The third symbol denotes for calling the emergency services if the patient suffers pain or any sort of seizure. The right-side cell icon is related to the necessity of the user to go to the toilet due to a motor disability. Each target was gazed at for 40 seconds.

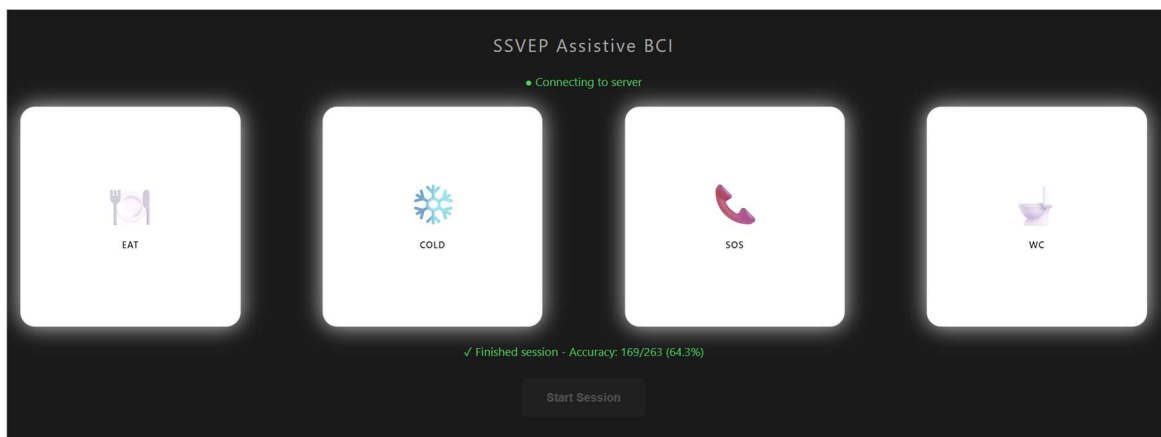


Figure 23 Online User Interface visualization.

Each session evaluated a target symbol at a time when the UI Start Session button was pressed, as observed at the bottom of the screen in **Figure 23**. The four cells flickered simultaneously at their fixed frequencies (8.57, 10, 12 and 15 Hz) throughout the session, and the user was asked to gaze at the previously defined target cell (see **Figure 24**). Next,

to analyze the procedure for other symbols, the target was changed and the session was repeated.

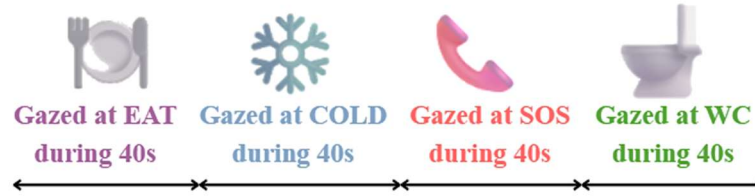


Figure 24 Recording online protocol scheme.

Throughout each 40 s recording iteration, the server displayed at the command-line terminal screen as displayed in **Figure 25**; the target assigned (Eat, Cold, SOS or WC), the achieved classification for that iteration, the four CCA correlations detected for each cell and the winning frequency which coincides with the maximal canonical correlation at each evaluation (see section 2.5.2 for better understanding).

```
C:\Users\carne\Desktop\mindaid-ssvep-bci\backend> python server.py
[Processing] Pre-computed filters:
- Bandpass: 7.0-70.0 Hz
- Notch comb: 50 / 100 / 150 Hz (Q=30)
- CCA harmonics: [1, 2, 3]
=====
SSVEP Online Assistive BCI
=====
One 40 s block per session: user gazes at the target symbol.
Classification: CCA winner on every 4s window.
Frequencies:
1. 🍴 Eat → 8.57 Hz
2. ❄️ Cold → 10.0 Hz target!
3. 📞 SOS → 12.0 Hz
4. 🚽 WC → 15.0 Hz
Trial duration: 40s
Mode: HARDWARE (Cyton board)
=====
✓ Cyton connected at COM5
Waiting 4s to fill the EEG buffer ...
Ready! Open the browser and click 'Start Session'.

✓ Client connected: ('::1', 64530, 0, 0)
[Recorder] Recording to: recordings\online_Cold_20260903_135418.txt
[CCA] 8.57Hz: 0.1557
[CCA] 10.0Hz: 0.2093
[CCA] 12.0Hz: 0.2024
[CCA] 15.0Hz: 0.1157
[Result] ❄️ Cold: Corr=0.2093 - ✅ CORRECT
[Time elapsed] 4.1620 (s)
[SUMMARY] Accuracy: 1/1 (100.0%)
[CCA] 8.57Hz: 0.1619
[CCA] 10.0Hz: 0.1871
[CCA] 12.0Hz: 0.2178
[CCA] 15.0Hz: 0.1333
[Result] ❄️ Cold: Corr=0.2178 - ❌ INCORRECT (detected 12.00Hz)
[Time elapsed] 0.0364 (s)
[SUMMARY] Accuracy: 1/2 (50.0%)
```

Figure 25 Server console commands visualization.

2.5.2 Python Data Analysis

In the real-time system, the classification was carried out continuously in the Python backend. Every 0.1 s, the most recent 4 s consecutive windows of EEG were taken from the data stream and preprocessed with the same pipeline described in section 2.4.3.

For each window, the Canonical Correlation Analysis was computed between the EEG signals recorded by the used channels (P7, P8, O1 and O2) and the reference signals of the four candidate frequencies (8.57, 10, 12 and 15 Hz), obtaining the first canonical correlation coefficient per candidate. The frequency with the highest correlation was taken as the predicted target for that window. The four correlation values, the winning frequency and the resulting prediction were displayed on the server console and sent to the user interface. A prediction was considered correct when the detected frequency matched the one assigned to the expected gazed cell.

It did not require training procedures, lowering its complexity. Up to three harmonics per frequency were contemplated, as done by Chen et al. [14], such that a neurological response was elicited for several frequencies. The used canonical component was equal to 1. If the canonical correlation value was maximum with respect to the other possible monitor interface frequencies at each analyzed window, the desired cell target was displayed as green, indicating an adequate selection.

The accuracy of the system was then computed every 40 s with the target previously assigned. The calculation of the accuracy value was defined as the proportion of correctly classified windows: the number of windows whose predicted target matched the assigned cell, divided by the total number of windows analyzed during the recording session. A window was counted as correct when its detected frequency matched the one assigned to the gazed cell (within a 0.5 Hz tolerance). This metric was updated and displayed continuously so that the accuracy obtained in real time from the data stream of predictions could be monitored throughout each recording.

2.5.3 Real-time Web Communication

Offline implementation (see section 2.4) split functionalities such that the software application gathered at the Python server consisted of the raw EEG data

acquisition and related to web UI visualization, and later through Jupyter Notebook scripts, the information was filtered, visualized and classified.

The real-time system implemented a Python backend server format with more modules, a WebSocket connected to an HTML frontend, which was responsible for the display of the flickering engine visual stimulation, real-time oscillatory visualization and output feedback classification set at the UI. These software element communication relations are seen in **Figure 26**.

The backend oversaw enabling data streaming connection with the Cyton board through BrainFlow libraries, preprocessed the entry EEG signal, classified the assigned frequency cells to the user's intention by its neurological response to the visual stimuli and sent the decision to the UI in order to show the results, as it is observed in the Python area depicted in **Figure 26**.

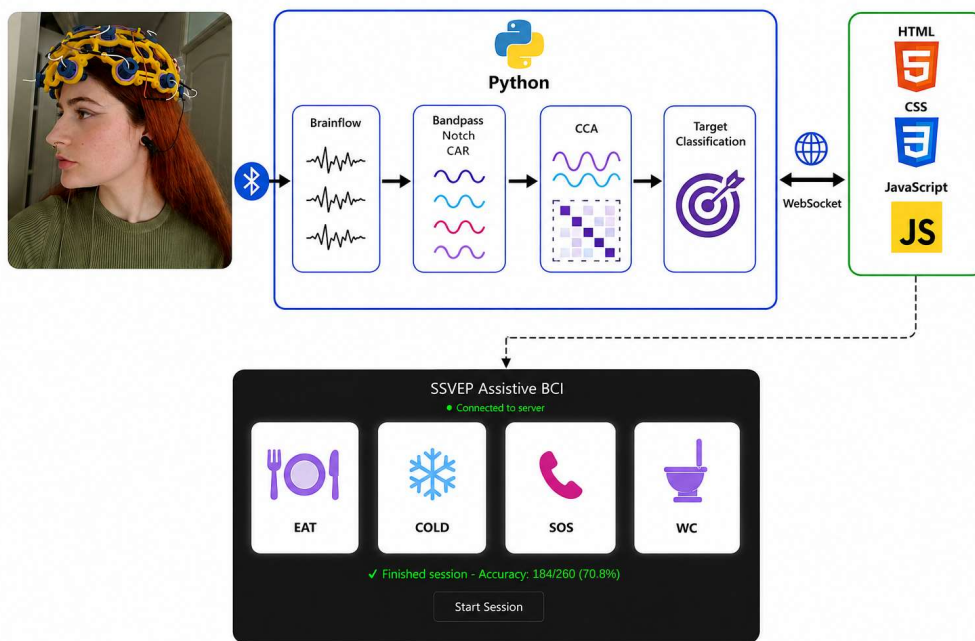


Figure 26 Assistive SSVEP BCI software system scheme (AI generated image).

The WebSocket was gathered as a bidirectional connection channel between the server and browser, transferring continuous data. According to certain parameters such as the window size acquisition, the user's fixation time and the estimated target cell, the server analyzed the signal, preprocessed it, classified it and sent the correlation results and the accuracy calculation to the UI.

3 RESULTS

3.1 Identification of optimal acquisition conditions

EEG signal quality and acquisition aspects were evaluated under different protocol conditions to ensure optimal performance of the system. The following **Table 3** shows the main evaluated conditions conducted through preliminary project characterization and the observations documented with respect to the effect of each of them upon the signal quality.

Table 3 Preliminary acquisition conditions evaluated and their impact on signal quality

Condition	Observations
No prior hair wash	High impedance values
Shampooing once	Moderate decrease in impedance values
Shampooing twice	Substantial decrease in impedance values
Braided hair (before the recording)	Facilitates electrode-skin contact due to hair being more compressed and separated
Day of recording (with respect to hair washing)	Higher signal quality when recordings are performed on the same day of hair washing
Morning vs. Evening acquisition	Differences occur due to user fatigue (higher before dinner time)
Electrode cleaning with a humid towel	Similar impedance values to those when no electrode cleaning is performed
Reference electrode position	Better impedance values when the reference electrode is a left ear clip than when it is an adhesive electrode on one or both mastoid bones

Central channels had a tendency to rail due to the dimensions of the helmet, which fitted closer to the frontal and occipital areas. Shampooing the hair twice and braiding as observed in **Table 3**, allowed central electrodes to be closer to the scalp, improving the

impedance values along all the channels used. Furthermore, it was not possible to accomplish registrations at night, as lamps or any lighting device caused signal interference coupling. The same event occurred when the charger was plugged into the laptop; or any Wi-Fi device, router or phone was close to the laptop during recording sessions.

Signal quality was also maximized when initial actions such as placing the headset and waiting for at least one minute to stabilize the electrode-to-skin impedances and adjusting all the electrodes properly until their impedance was below $750\text{ K}\Omega$ were taken, which typically took from 15 to 30 minutes. Under these optimal experimental conditions, measured impedance values typically ranged from $20\text{ K}\Omega$ to $600\text{ K}\Omega$, as shown in **Figure 27**.



Figure 27 Impedance values checked before attempting a new recording session.

To sum up, the maximum optimal signal quality acquisition was achieved in subsequent recordings when: 1) the hair was braided the day before, 2) the hair was shampooed twice the same day of acquisition, 3) the EEG acquisition was performed during the day and especially during the morning, 4) recordings were performed away from any possible electrical interference (lighting or electronic devices).

3.2 Preliminary EEG responsiveness results

Figures 28 and 29 show the results of the Head Plot and Focus Widgets during the preliminary EEG responsiveness tests. It can be seen that the Head Plot Widget offered neurological response regarding visual stimulation among occipital (red color appearance) and frontal areas (blue color appearance), as observed in Figure 28; and the Focus Widget displayed that the audio turned on changed in volume during the session. Especially on channel 7, variations in shape and amplitude were observed, as seen in the red channel in Figure 29. Both tools among relaxation tasks (EC) or focusing ones (EO) provided neurological feedback.

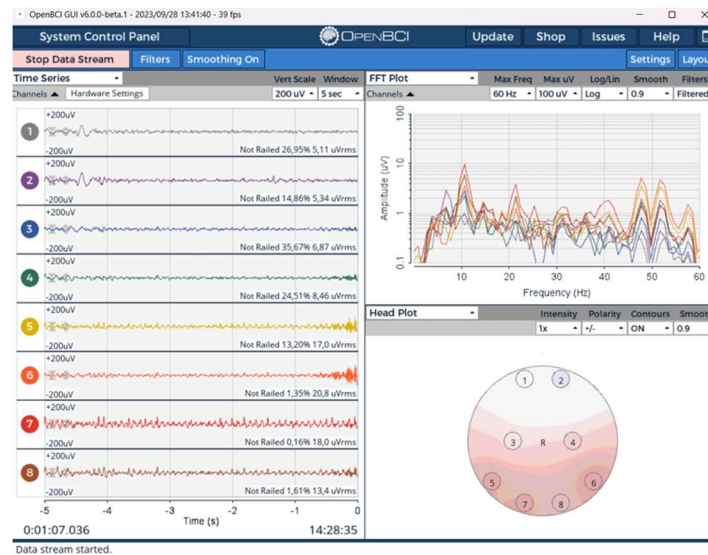


Figure 28 Head Plot Widget during preliminary EEG responsiveness tests.



Figure 29 Focus GUI Widget during preliminary EEG responsiveness tests.

Figure 30 shows the channel-averaged spectrogram of the EEG signals recorded during the Eyes Open and Eyes Closed (EC) condition trials in logarithmic frequency scale representation. As it shows, the beep sound emitted indicating the user to switch to the next event was observed as a neurological response in the spectrogram at the beginning or end of the second and fourth epochs. Other events were also recognized, such as blinking, as seen in the first and third epochs in the following **Figure 30**. Swallowing saliva, blinking several consecutive times or changing breathing patterns introduced noise in the registrations.

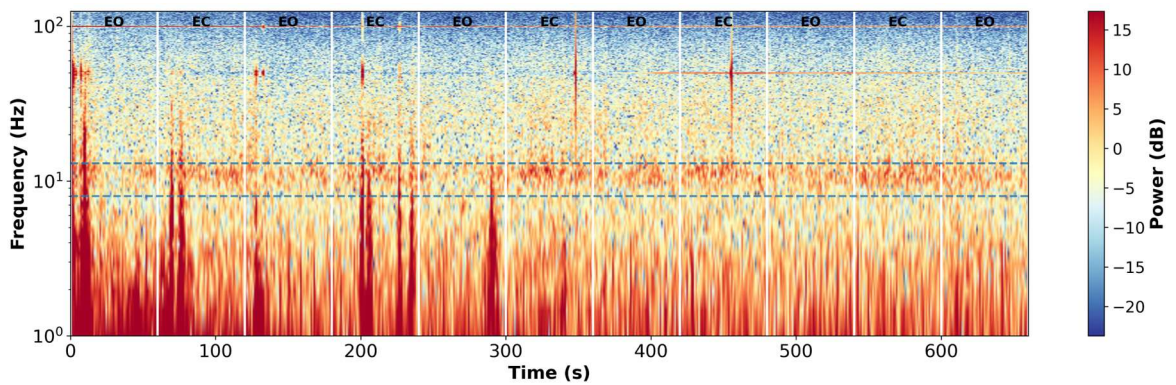


Figure 30 Averaged Spectrogram depicting EC vs EO alternation.

Figure 31 shows the relative energy per frequency band of the averaged EEG data over all valid channels and applying a 0.5-45 Hz bandpass filter. EC epochs are represented as circles while EO epochs are marked as diamonds. As it shows, EC scenarios were associated with higher Alpha frequency band activity than EO scenarios, which is observed in the light blue boxplots.

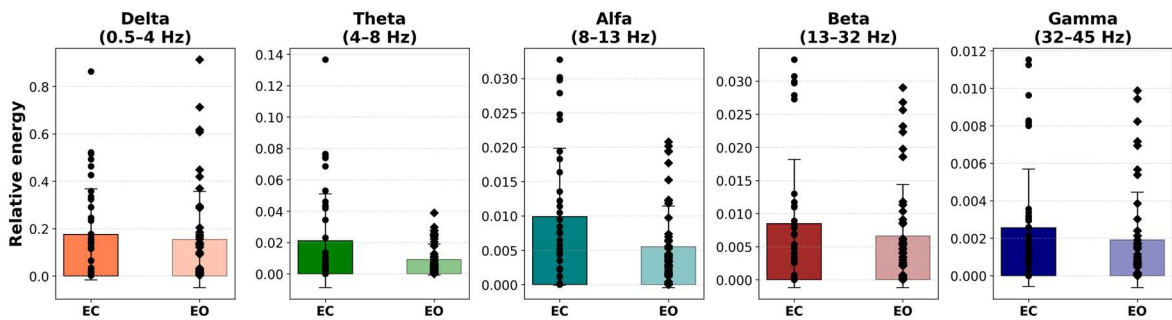


Figure 31 Relative energy per frequency band boxplots in eyes-closed (EC, circles) vs eyes-open (EO, diamonds) epochs.

3.3 BCI Offline Assessment

Figure 32 represents the EEG signal in the temporal domain after applying a double band-pass (7-70 Hz) and comb preprocessing filters. The response of the four processed channels (P7, P8, O1 and O2) is analyzed while the user gazed at the 12 Hz target, with the four cells arranged in a row layout and flickering simultaneously on the screen. As shown, blinking occurred during the recording, which resulted in artifacts in temporal sequences seen in blue square marks. Testing periods lasted 160 seconds for this row offline implementation, where the user presented involuntary neck movements that produced EMG artifacts as seen in **Figure 32**, depicted with light green color rectangle shapes.

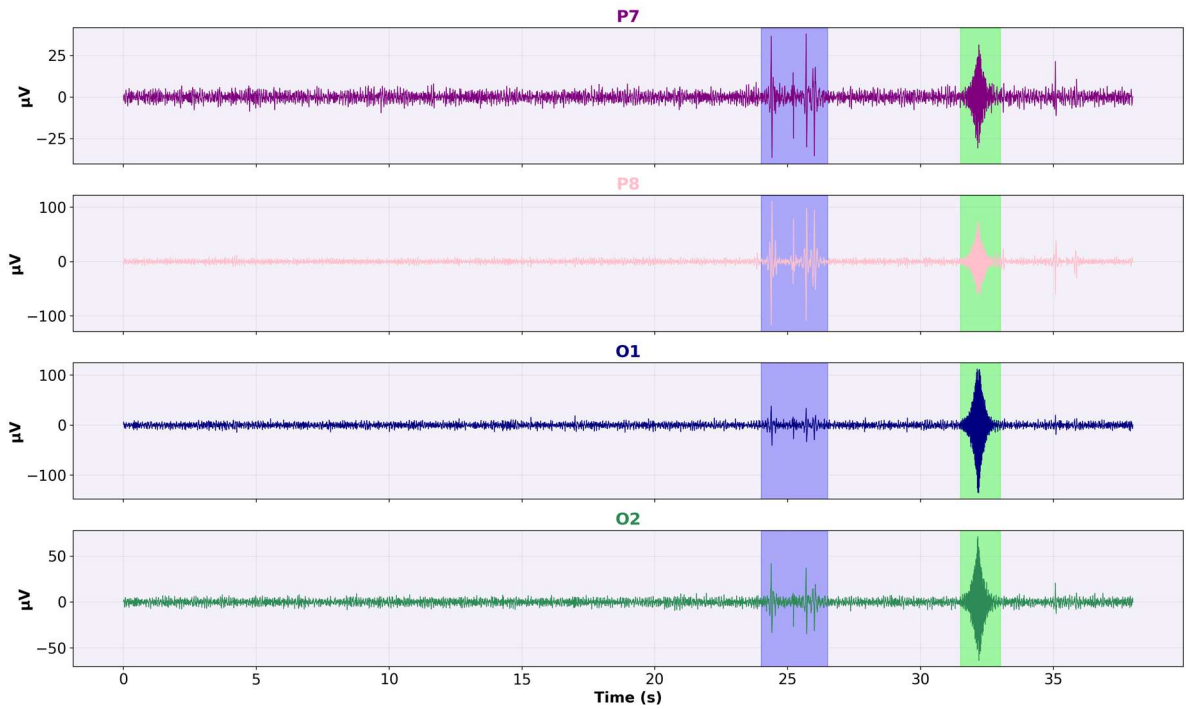


Figure 32 EEG Temporal Domain Signal of 12 Hz for row-shaped offline analysis.

Figure 33 shows the PSD at the three first harmonics for the row-shaped arrangement. For each processed channel (P7, P8, O1 and O2), the spectral energy response at 10 Hz and its first three following harmonics is represented, comparing the energy obtained for the target frequency with the remaining candidates not seen as prominent peak responses (8.57, 12, 15 and their harmonics). As observed in the following PSD representation in **Figure 33**, it is confirmed that the user generated a neurological response to the target flickering frequency and their harmonics, seen as

prominent PSD peaks at 10 Hz (fundamental frequency), 20 Hz (second harmonic), 30 Hz (third harmonic) and 40 Hz (fourth harmonic) values. This was also observed in the PSDs of the EEG signals recorded when the other target flickering frequencies (8.57, 12 and 15 Hz) were tested in **Appendix 2, Figures A.2.1-A.2.3**.

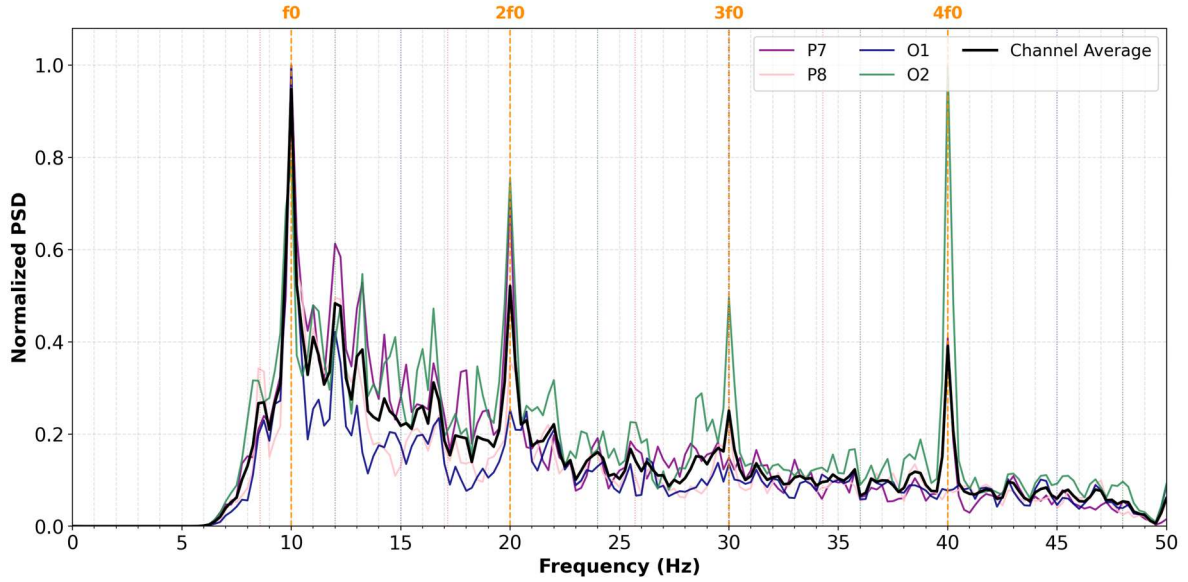


Figure 33 PSD at the fundamental and following three harmonics of 10 Hz (user interface: row-shaped arrangement).

Figures 34 and 35 show the summed-PSD feature extraction of the 10 Hz target for the two interface layouts evaluated: the row-shaped arrangement in **Figure 34** and the cross-shaped arrangement in **Figure 35**. In both representations, the horizontal axis lists the four candidate frequencies (8.57, 10, 12 and 15 Hz) and the vertical axis represents the summed spectral energy over the target frequency and its first two harmonics (f , $2f$ and $3f$). The gazed-at target of 10 Hz is displayed in occipital channels from both layouts in red, so that its prominence relative to the remaining candidates can be assessed, especially comparing parietal channels.

Figure 34 shows the total PSD summation representation for each candidate frequency for all processed channels. The expected signal coincides with the registered one, as observed as prominent in comparison with other energy frequency values (8.57, 12 and 15 Hz). Registrations for other matching gazed-at cells with their corresponding candidate frequencies are gathered in **Appendix 2, Figures A.2.4-A.2.6**

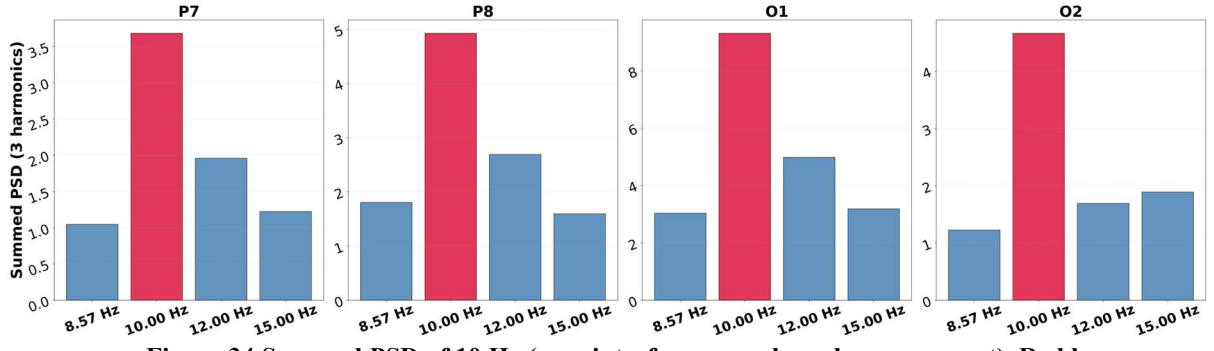


Figure 34 Summed PSD of 10 Hz (user interface: row-shaped arrangement). Red bar: target frequency.

Cross-shaped offline layout implementation offered misclassifications and not enough discrimination at summed PSD energy classification, as seen in **Figure 35**.

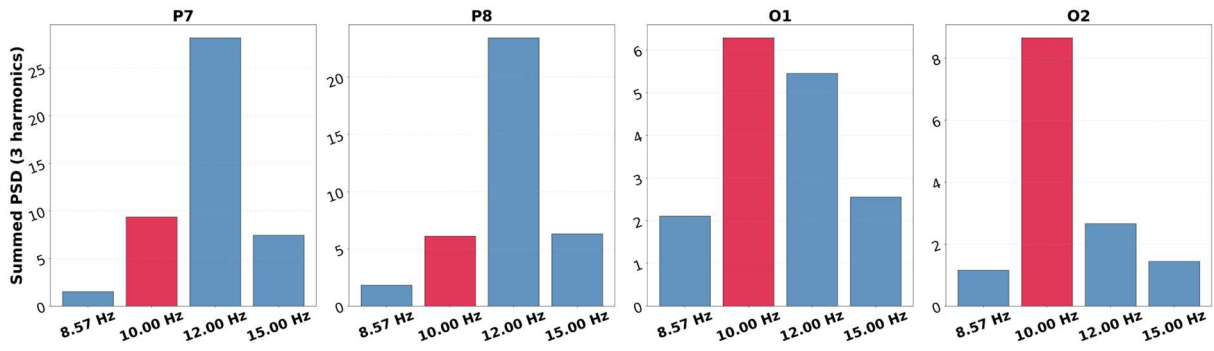


Figure 35 Summed PSD of 10 Hz (user interface: cross-shaped arrangement). Red bar: target frequency.

Figures 36 and 37 show the Canonical Correlation Analysis of the 10 Hz target for the two layouts: the row-shaped arrangement in **Figure 36** and the cross-shaped displacement in **Figure 37**. In both plots, the horizontal axis lists the four candidate frequencies, and the vertical axis represents the first maximum canonical correlation obtained when the reference signals were built from the target frequency and its first two harmonics (f , $2f$ and $3f$). The bar associated with the gazed-at target is displayed in red.

In the CCA classification in **Figure 36**, it is observed that the discrimination of the target value linked to 10 Hz oscillates between 0.47 and 0.56 as maximal first canonical correlation values, gathering small false positive values. Minimum correlation values at f_0 (fundamental frequency) is of 0.21 and at subsequent frequencies of 0.32 ($2f_0$) and 0.36 ($3f_0$). CCA classification gathers higher correlations with three harmonics implementation (0.56) in comparison with the fundamental frequency alone (0.47)

application as observed in **Figure 36**. Other CCA discriminations are gathered in **Appendix 2, Figures A.2.7-A.2.9**.

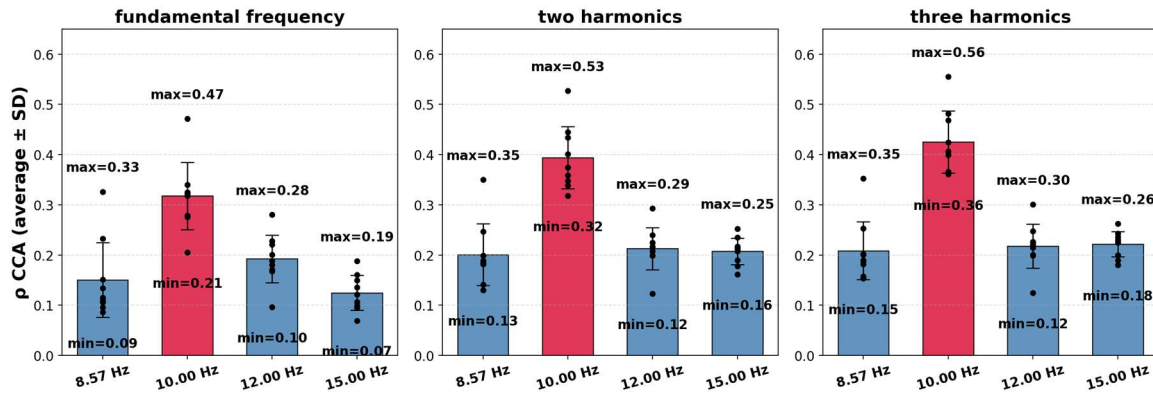


Figure 36 CCA at the first three harmonics of 10 Hz (user interface: row-shaped arrangement). Red bar: target frequency.

The other considered offline implementation allowed CCA 10 Hz target classification with maximum values oscillating from 0.60 to 0.66, as seen in **Figure 37**. Minimum correlation values range from 0.38 to 0.46, as seen in **Figure 37**. Higher correlation values increase as a higher number of harmonics is considered, as happened with the row implementation.

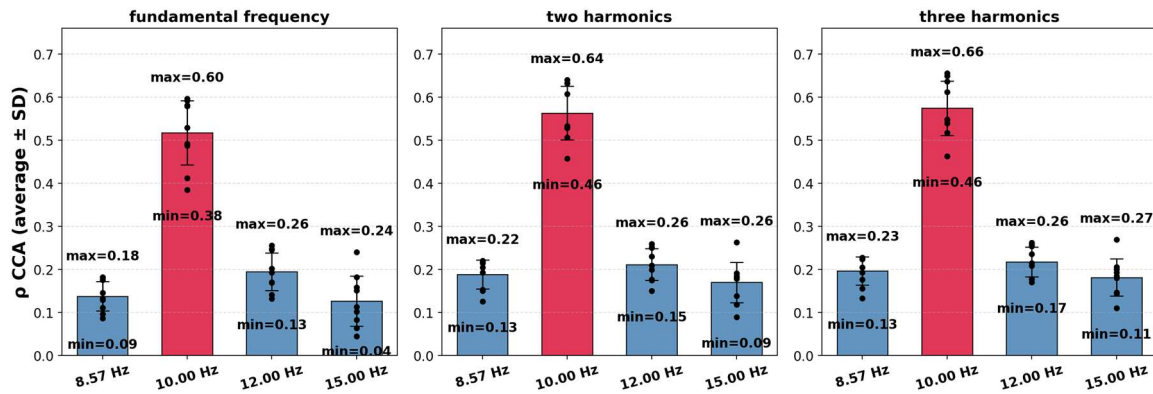


Figure 37 CCA at the first three harmonics of 10 Hz (user interface: cross-shaped arrangement). Red bar: target frequency.

The cross-shape offline implementation considered depicted muscular artifacts that were seen at temporal graphs similar to the ones observed in **Figure 32**, while attempting to look at each cell from up to left, then to right and finally downwards. These artifacts also happened with the row displacement. However, in the cross-shaped layout, it occurred more often. Either PSD or CCA classifications allowed target frequency

discrimination, but despite providing higher canonical correlation values, the summed PSD produced misclassifications as observed in **Figure 35**. Therefore, the row displacement was considered for the online implementation detailed in the next section.

3.4 Online BCI System Assessment

Two subjects tested the online version of the assistive SSVEP BCI system. The first subject had great hair volume and presented difficulties fixating at the target cell while maintaining a relaxation state at the same time. The accuracy of these registrations oscillated between 55%-71% as seen in **Figure 38**. The other subject, who presented less hair volume and focus capability, achieved an accuracy range between 65%-88% as seen in **Figure 39**.

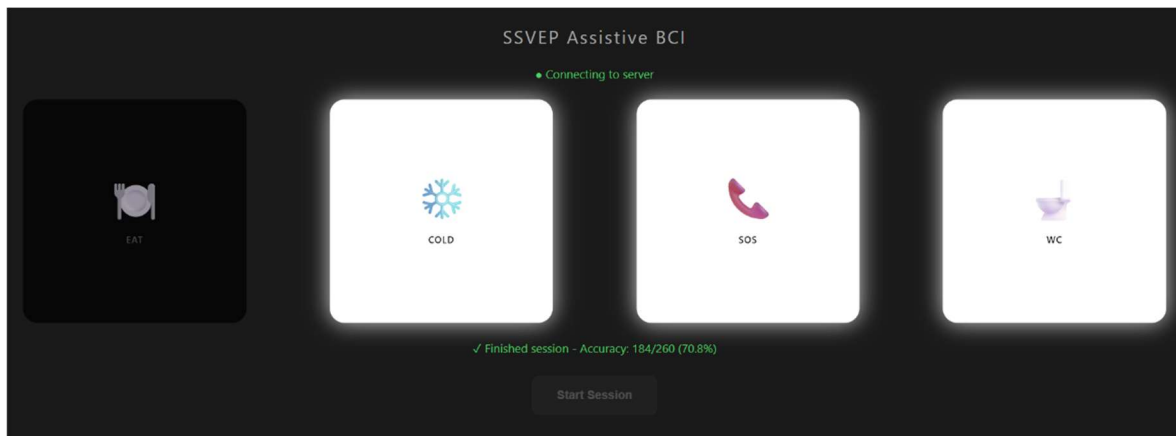


Figure 38 Subject 1 end-of-session UI visualization.

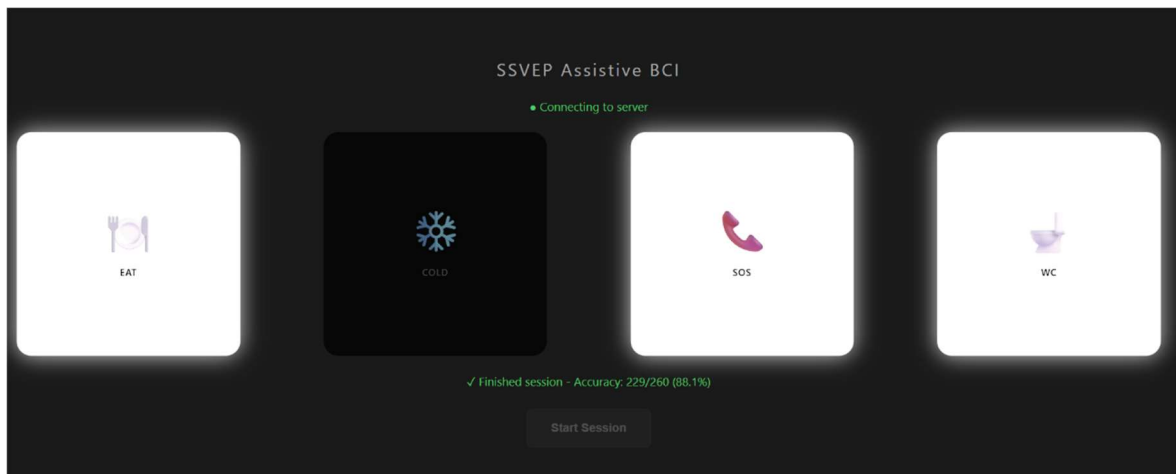


Figure 39 Subject 2 UI finished recording session.

The processing and classification latency per window was estimated from the elapsed time reported by the server console for each real-time window processed, excluding the initial buffer filling step of 4 seconds for Cyton board connection. The average latency was very similar for both subjects, 29.7 ms (median 31.6 ms, SD 5.1 ms) for Subject 1 and 30.8 ms (median 31.9 ms, SD 6.0 ms) for Subject 2.

Table 4 depicts accuracy values averaging 62.9%, considering that 1040 data evaluations were performed by Subject 1 during recording sessions for each icon target. The first canonical correlation values tended to be around 0.220 $\pm$ 0.059 with a maximal correlation value with respect to all targets of 0.426 and a minimal of 0.084, as observed at the end of the last column of **Table 4**.

Table 4 Accuracy results of Subject 1 from real-time recording trials.

Cell	Frequency (Hz)	Accuracy	Average ρ	Range ρ
Eat	8.57	160/260 (61.5%)	0.192 $\pm$ 0.022	[0.148, 0.240]
Cold	10.00	169/263 (64.3%)	0.225 $\pm$ 0.044	[0.140, 0.328]
SOS	12.00	184/260 (70.8%)	0.234 $\pm$ 0.067	[0.114, 0.416]
WC	15.00	141/257 (54.9%)	0.227 $\pm$ 0.078	[0.084, 0.426]
TOTAL	-	654/1040 (62.9%)	0.220 $\pm$ 0.059	[0.084, 0.426]

Table 5 displays the same parameters as **Table 4** for Subject 2, resulting from each analyzed processed window such that 1034 evaluations were registered, with an accuracy rate of 74.6%, average first canonical correlation value of 0.327 $\pm$ 0.111 and maximum and minimum values across all sessions of 0.645 and 0.095, respectively.

Table 5 Accuracy results of Subject 2 from real-time recording trials.

Cell	Frequency (Hz)	Accuracy	Average ρ	Range ρ
Eat	8.57	229/260 (88.1%)	0.309 +- 0.096	[0.095, 0.567]
Cold	10.00	179/253 (70.8%)	0.311 +- 0.065	[0.165, 0.432]
SOS	12.00	192/259 (74.1%)	0.385 +- 0.149	[0.134, 0.645]
WC	15.00	171/262 (65.3%)	0.301 +- 0.096	[0.130, 0.469]
TOTAL	-	771/1034 (74.6%)	0.327 +- 0.111	[0.095, 0.645]

Figures 40 and 41 show the confusion matrices of the classification results performed for each subject. As can be seen, selections that fall along the main diagonal correspond to correct classifications, while the other elements indicate the misclassified targets selections.

Misclassifications tended to occur between consecutive cells on the screen monitor as observed in both confusion matrices, except in Subject 2 with the Eat (8.57 Hz) icon, which was not discriminated against the next cell and instead was selected as the SOS symbol (12 Hz), as observed in the first row and third column in **Figure 41**.

Particularly, Subject 2 registers Eat (8.57 Hz) as the highest value while Subject 1 corresponds to SOS (12 Hz) as its highest accurate rate symbol registered, as seen in the third row and column in **Figure 40**. Registrations regarding Cold (10 Hz) and SOS (12 Hz) targets gained similar accuracy values, with a difference of 10 correct evaluations between subjects for the Cold icon and in the case of the SOS symbol with a difference of 8 selections. Also, in both subjects, the WC (15 Hz) target is the least accurate registered symbol, compared to the number of evaluations performed in their respective recording sessions with respect to the other candidate frequencies.

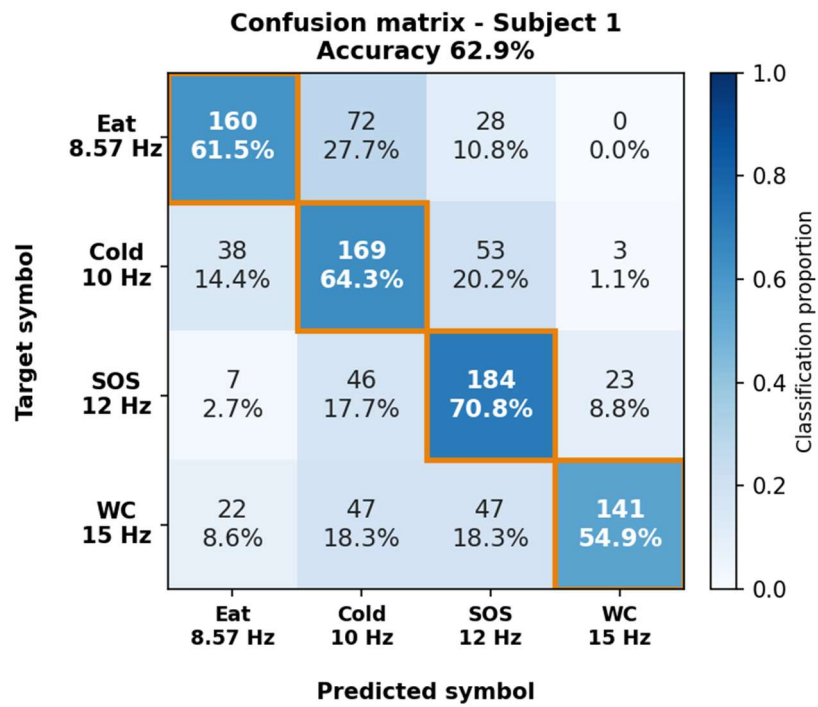


Figure 40 Subject 1 confusion matrix.

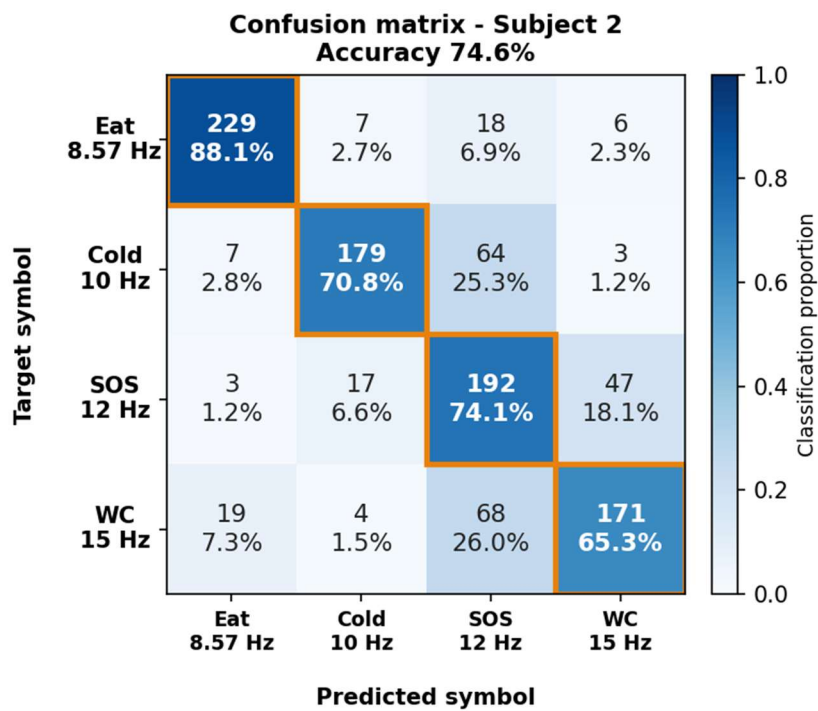


Figure 41 Subject 2 confusion matrix.

4 DISCUSSION

The Assistive SSVEP BCI system built for this Bachelor Thesis is aimed at patients who have lost communication abilities and need an external tool to indicate their desired assistive functional intentions by using EEG signals. The project application was inspired by the BCI developed by Dehzangi et. al and Farooq [27] and based on the visual cortex neurological modulation phenomenon [4]. The system is a low-cost, dry-electrode, open-source, real-time BCI prototype developed using the available OpenBCI hardware equipment. The implementation achieved represents an exogenous (flickering), synchronous (specific gazing time) approach, easy to build, with a low number of electrodes (P7, P8, O1 and O2) and no training requirement. The prototype design and its intended application were decided by considering the hardware limitations, the complexity of implementation, the accuracy and the selection time. Particularly, the decision to implement the system on a laptop with limited resources restricted the type of BCI application that could be developed, since the frequencies of the available cells were determined by the monitor refresh rate available. Due to the complexity and time constraints of the project, parts of its design were simplified until a functional low-cost system was obtained.

The BCI system was developed in successive phases: 1) acquaintance with the OpenBCI hardware and software; 2) optimization of the acquisition protocol and preliminary EEG responsiveness trials; 3) offline BCI system implementation, built on existing compatible repositories (see **Appendix 1**), and evaluation; and 4) real-time BCI implementation and assessment of its classification performance. In the offline stage of the study, signal preprocessing and classification steps were performed separately from the EEG data acquisition and recording, which reduced the configuration complexity during recording sessions before implementing the online BCI system and allowed optimizing relevant aspects such as the arrangement of the user interface cells or the signal processing pipeline that maximized the detection of the target flickering frequency. Once consistent results were obtained via offline implementations, a posterior online format was contemplated.

4.1 Evolution of the BCI system prototype

Before arriving at the final design, several prototypes of the system were built and tested under different conditions. Different visual stimuli and display configurations were tried, and their impact on SSVEP detection was assessed throughout the process. Several factors influenced performance. Stimuli on the user interface were presented as white flickers on a dark background, which have been demonstrated to give the best response [28], and a user-to-screen distance of 40-60 cm was optimal. Only four parietal-occipital channels were used, since frontal channels spread ocular artifacts (EOG) through CAR when they were included in the analysis and worsened classification; CAR was therefore beneficial only with this reduced setup approach. Both CAR and the use of three harmonics (rather than the fundamental alone) improved PSD feature extraction analysis and CCA target classification.

As expected for a visual BCI paradigm, the occipital electrodes showed the greatest EEG responsiveness to visual stimulation. The relative-energy analysis performed with the EEG signals from the preliminary tests regarding EO vs EC conditions reproduced the classic Berger effect (higher Alpha activity during the Eyes Closed) [10], confirming that the expected effect of visual stimulation on the EEG was captured.

Also, during the development of the system, several classification criteria were tested and refined. The best results were obtained when the target was selected as the frequency with the highest canonical correlation coefficient among the four candidate cells within each recorded window, rather than using fixed thresholds, which is the criterion finally adopted for the real-time system.

Overall, these conducted tests confirmed that the EEG acquisition system employed was appropriate for developing a functional SSVEP-based BCI.

4.2 Classification Performance of the Online BCI System

Although the BCI system prototype was modified over the project development to maximize its flickering frequency detection performance, the average accuracy value across them was 68.7 % (ranging from the lowest value of 54.9% to the highest of about 88.1% across targets and subjects). The accuracy obtained for this developed system is

lower than the values frequently reported in other SSVEP BCIs, which often exceed an accuracy of 90% [14,15]. These moderate performance values could have been influenced by different factors that are described next.

First, dry electrodes were chosen for their short preparation time and due to utility and conformity reasons, at the expense of higher contact impedance (decrease of scalp-electrode conductivity) and greater sensitivity to interference (Wi-Fi, phones, lighting, chargers, fans and air conditioning). This scenario consequently provided a low SNR entry signal and wider EEG signal acquisition variability across trials and subjects that strongly affected the information transfer rate per iteration. Furthermore, while blinking produced only brief, punctual artifacts that did not compromise classification, muscular activity, neck fatigue, longer fixation times and mental effort degraded the signal more noticeably. These factors, together with laptop constraints (RAM and refresh rate), may have limited the achievable performance of the implemented BCI system.

Other factors that influenced the system's classification performance were the window size for acquisition and the amount of time by which the user had to gaze at the target, as higher fixation time introduced more artifacts due to the user's tiredness appearance. Furthermore, signal windows were processed as non-overlapping segments in the offline system and as consecutive overlapping windows in real time, which increased the online temporal resolution at the cost of higher variability. It is also worth noting that the offline BCI version, which assessed stored recordings, classified target frequencies more accurately than the real-time system due to processing data sequentially, analyzing the whole loaded file. In contrast, the online version introduced a slight latency across its communication modules (filtering, WebSocket exchange and UI updates) and needed to process and update a continuous data stream at each iteration, demanding powerful performance on a restricted laptop. These latency values depend not only on the processing pipeline and the laptop's performance, but also on the subject's ability to maintain visual focus.

Regarding the target performance, both subjects reported that the Cold (10 Hz) and SOS (12 Hz) targets were easier and faster to select and sustain over time, which was reflected in similar accuracy values for these two cells. In contrast, the Eat (8.57 Hz) and SOS (12 Hz) targets behaved differently between subjects: one subject obtained the

highest accuracy for Eat, whereas the other did so for SOS. The WC target (15 Hz) was consistently the least accurate for both subjects, suggesting that the highest stimulation frequency lies outside the Alpha frequency band and was the most difficult to detect reliably with this setup.

Overall, these results are consistent with the literature, which reports better SSVEP responses when the stimuli are located closer to the center of the visual field [4], and point to the stimulation frequency and the target position as relevant factors affecting the classification performance.

Despite this, as discussed above, accuracy remained above 60% and target selection was always achievable in all recording sessions. For an assistive communication context, where a reduced set of vital needs must be selected without any physical interaction, this level of performance can be considered promising proof of concept, although further optimization would be required before its use in a real clinical application.

4.3 Limitations and future lines

Apart from the technical and environmental factors mentioned that could have influenced BCI's performance, other aspects could have also affected the detection of the target flickering frequency. Particularly, both subjects felt tiredness when they tested the online BCI and attempted consecutive trials. Therefore, the system performance could probably be improved in future studies by allowing more rest between trials, using a comfortable chair (to rest the back and neck) and shortening the gazing time, since longer recordings did not necessarily increase accuracy.

The design of the different BCI modules was progressively modified in order to maximize the system's performance. In addition to changing the cell layout from a cross to a row arrangement, several preliminary tests were carried out that were not fully reported in the Results section, such as evaluating different cell colors, title visualizations, feedback visualization, distance between cells and flickering background combinations, from which the white-to-dark configuration provided the best response. A greater number of cells were attempted to be implemented, but the user's responses did not provide

enough accuracy and were often misclassified by consecutive cells. A solution that could be explored for future implementations could be to introduce more cells with those candidate frequencies but with variations in their phases as depicted by scientific literature [29].

Another promising solution that could improve the detection of the flickering frequency in future projects could be the implementation of an amplitude-based rejection algorithm that discards contaminated signal windows before classification to remove strong muscular artifacts contaminating the signal severely. In addition, a correlation threshold could be applied so that signal windows whose canonical correlation value lay below it were not classified, reducing the number of erroneous frequency selections from the recordings. Further improvements might come from the use of a desktop computer with a powerful CPU and screen refresh rate, robustness classification algorithms, a user calibration stage prior to recordings, a greater number of electrodes or trainable models (like neural networks with cloud computing) rather than switching to a gel-based headset.

Overall, this thesis showed that an online assistive SSVEP BCI can be built with low-cost equipment. However, the developed system could be enhanced with further adjustments, given its moderate performance. Additionally, larger studies including more participants and sessions would also be required to assess the system's generalization and its performance for assistive applications.

5 CONCLUSIONS

This Bachelor's Thesis developed a real-time Brain-Computer Interface (BCI) based on Steady-State Visual Evoked Potentials (SSVEPs) to control a virtual assistive keypad using non-invasive EEG signals. The final system used four parietal and occipital channels (P7, P8, O1 and O2) and four simultaneously flickering targets at 8.57, 10, 12 and 15 Hz frequencies, each associated with a vital assistive action. Canonical Correlation Analysis (CCA) was implemented for target classification by using the fixed stimulation frequencies and their consecutive harmonics.

Offline analysis confirmed EEG responsiveness to visual stimulation at the different flickering frequencies and showed that including harmonics and filtering improved target discrimination. The online system was successfully evaluated with two participants across four sessions each, achieving an overall average accuracy of 68.7%, with individual target cell accuracies ranging from 54.9% to 88.1%. These results demonstrate the feasibility of detecting the user's desired targets in online operations and providing visual feedback through a low-cost system with a reduced number of EEG channels.

Performance was affected by misclassifications between neighboring targets, muscle and neck movement artifacts, visual fatigue, electrode contact, dry electrodes, electrical coupling interference and hardware and processing limitations. Despite these constraints, the project achieved its objective of integrating EEG acquisition, preprocessing, SSVEP detection, CCA classification, and real-time web User Interface feedback into a complete functional prototype.

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APPENDIX 1 - Open-source platforms considered

Several available open-source platforms (GitHub repositories) regarding BCI applications were considered. All of them comprehended EEG acquisition and signal processing pipelines as well as BCI system implementations. It was searched for OpenBCI Cyton hardware compatibility, Python-based processing implementation, support for the SSVEP paradigm and suitability for real-time performance by using parietal and occipital channels.

The tools provided at the following table indicate the characteristics of all the tools searched for BCI development availability. The ones described next were the sources considered for implementation due to their similarity and compatibility with the system developed.

BrainFlow is a multi-language platform that comprises a group of code libraries that contain universal and native compatibility with the OpenBCI Cyton. It was chosen as the EEG signal acquisition layer due to its easy application and because it enabled real-time data processing. This was achieved by calling the appropriate BrainFlow functions: filtering methods, windowing, downsampling, noise reduction, data streaming, analog-to-digital data conversion (microvolt to numpy data conversion), etc.

BCI_SSVEP and **t9_ssvep** were considered as references due to owning simple keyboards. Both repositories are structured such that they are modular and scalable, which were considered purposes. Modules from BCI_SSVEP, like `loadClassifier.py`, `preprocessing.py` or `flickit.py` (flickering engine), were taken for inspiration. T9_SSVEP contains a cell assignment through identifiers that will match the data that arrives via streaming, which served as an important concept of implementation for this project.

Bci-essentials-python was used as a reference because of switching from an online configuration and real-time operation to an offline one. It was possible to perform pipeline validation of the software (interface appearance and data exchange workflow) without performing real-world scenarios (i.e., requiring the hardware). This served as an inspiration for the two arranged modes: the one related to hardware utilization and the simulated demo, which refers to the synthetic data performance implemented at the online version of this thesis.

NeuroTechX and **WATOLINK** were attached to the official OpenBCI Documentation, using identical equipment and paradigm. Both repositories comprise a browser-based speller interface (virtual keyboard) and a Python-based server, where the acquisition, processing and classification of the EEG signals were achieved. Both also employ parietal and occipital electrodes. NeuroTechX served as an inspiration regarding the CCA classification algorithm, based on assigning oscillatory frequencies to the highest signal correlation. With respect to the second repository, the frontend structure was contemplated, especially the use of sockets.

Tool / Platform	OpenBCI Compatibility	SSVEP BCI Compatibility	Reference
BrainFlow	Yes (native support)	-	[30]
BCI_SSVEP (deebbolger)	Yes	Yes	[31]
T9_ssvep (ludovicdmt)	Yes	Yes	[32]
bci-essentials-python (kirtonBCIlab)	Yes (via LSL connection; OpenBCI has by default Serial)	Yes	[33]
NeuroTechX-McGill-2021	Yes	Yes	[34]
Mind-speech-interface-ssvep (WATOLINK)	Yes	Yes	[35]
uniBrainSpeller (Tsinghua University)	No	Yes	[36]
OpenViBE	No	Yes	[37]
MEDUSA	No (requires University of Valladolid biomedical student's license)	Yes	[38]
BciPy	Yes	No (P300) speller)	[39]
SSVEP_OpenBCI (jnaulty)	No (Python version deprecated)	Yes	[40]

APPENDIX 2 - Additional graphical representations

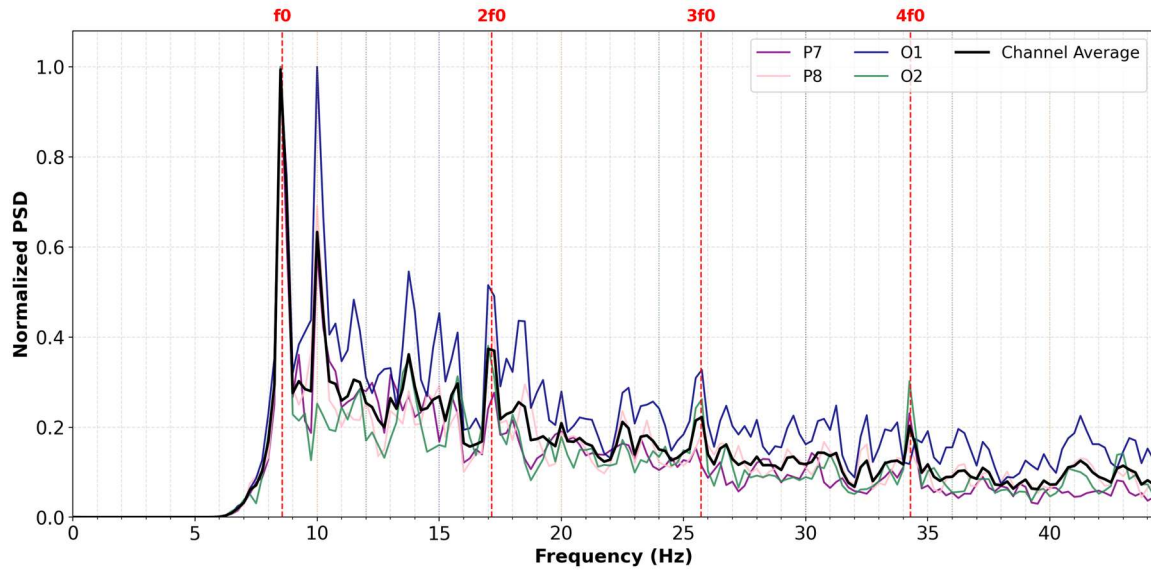


Figure A.2. 1 PSD at the first three harmonics of 8.57 Hz (user interface: row-shaped arrangement).

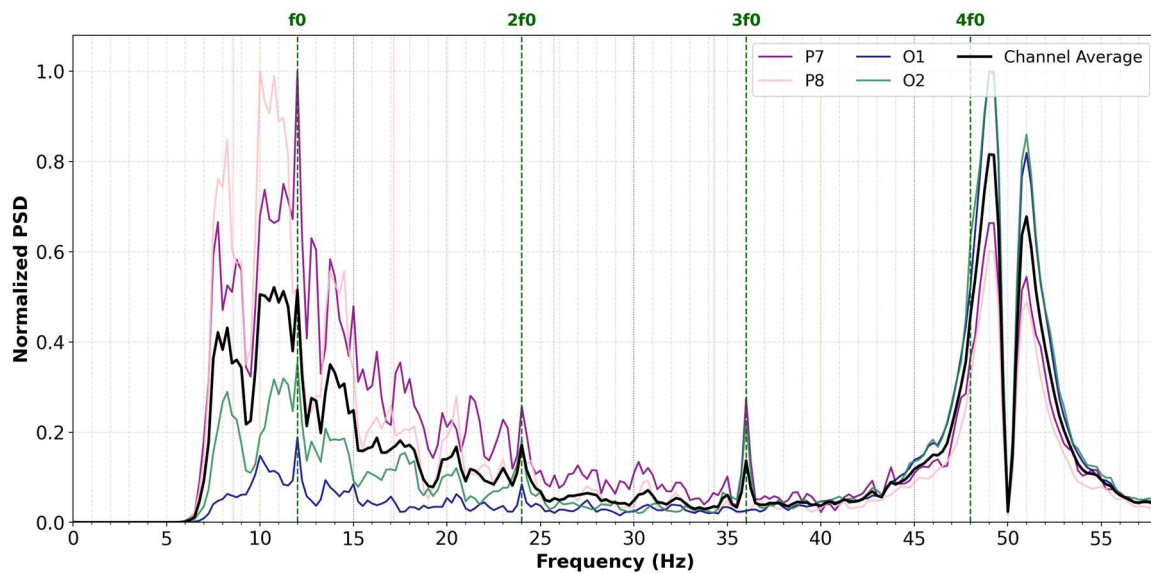


Figure A.2. 2 PSD at the first three harmonics of 12 Hz (user interface: row-shaped arrangement).

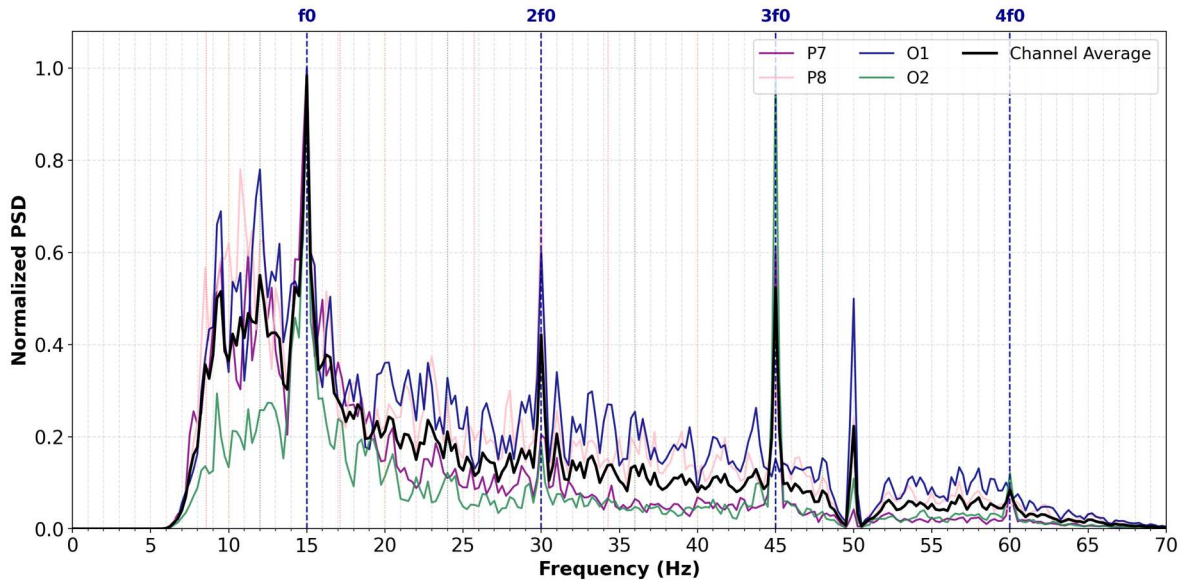


Figure A.2. 3 PSD at the first three harmonics of 15 Hz (user interface: row-shaped arrangement).

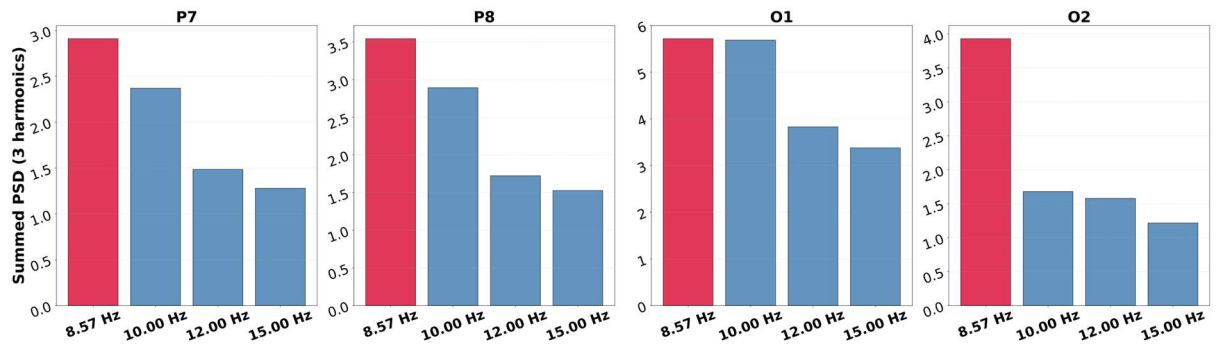


Figure A.2. 4 Summed PSD of 8.57 Hz (user interface: row-shaped arrangement). Red bar: target frequency.

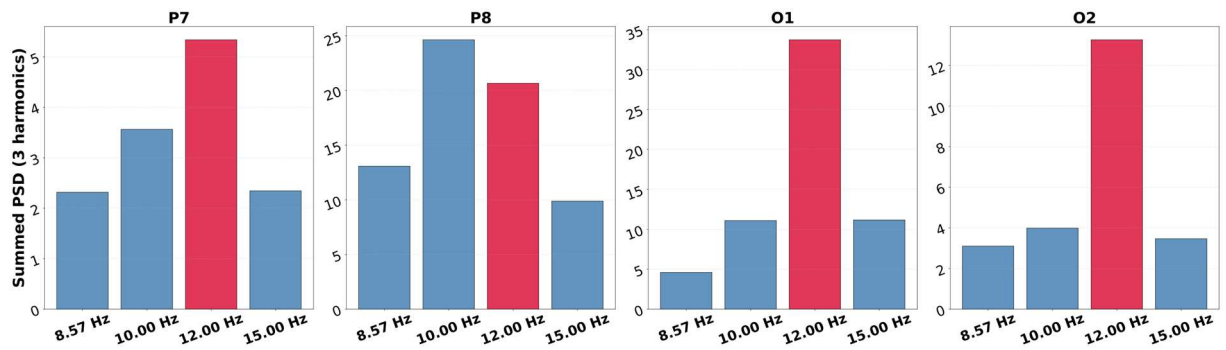


Figure A.2. 5 Summed PSD of 12 Hz (user interface: row-shaped arrangement). Red bar: target frequency.

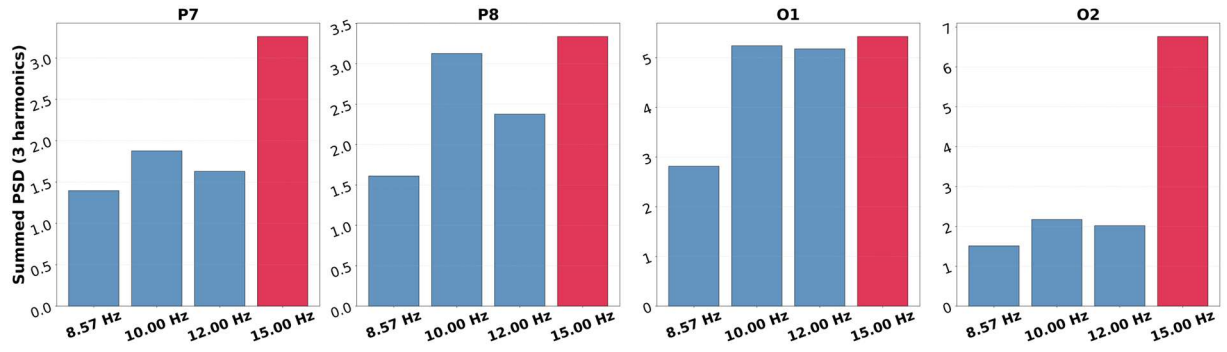


Figure A.2. 6 Summed PSD of 15 Hz (user interface: row-shaped arrangement). Red bar: target frequency.

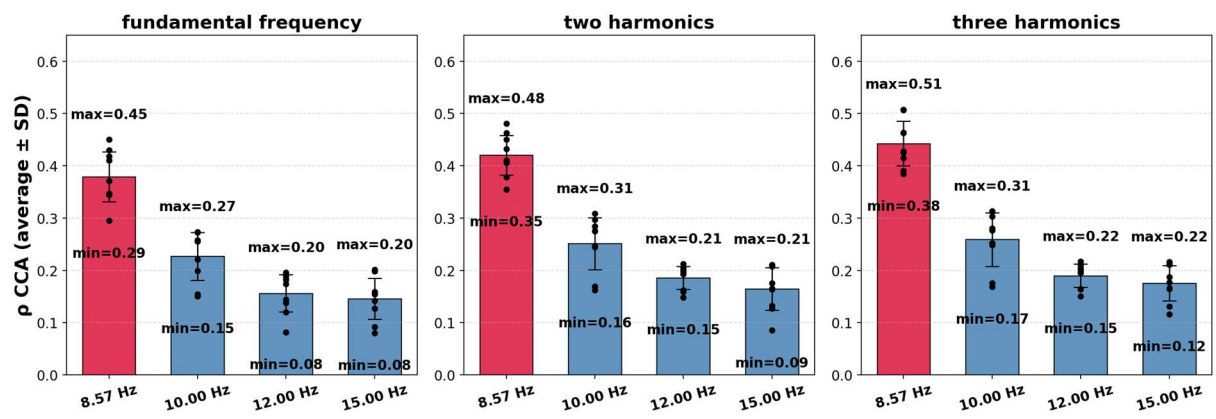


Figure A.2. 7 CCA at the first three harmonics of 8.57 Hz (user interface: row-shaped arrangement). Red bar: target frequency.

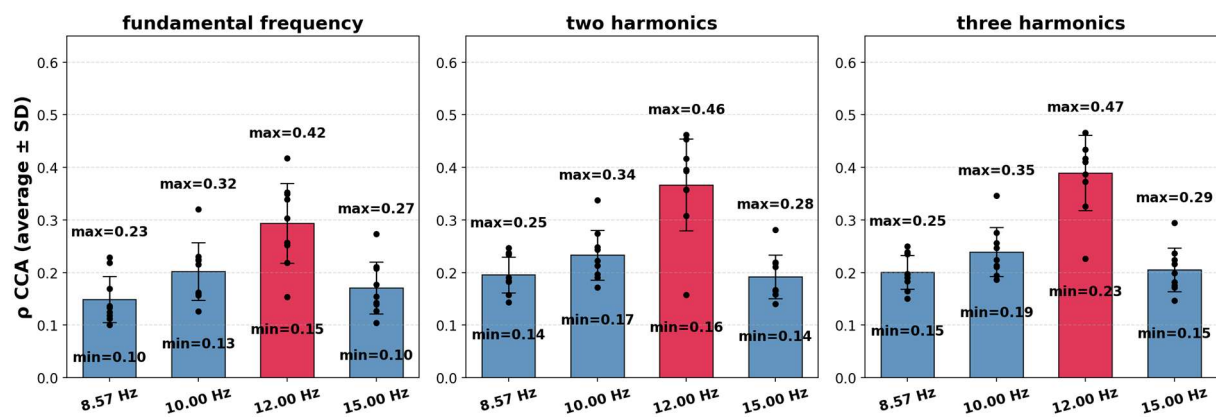


Figure A.2. 8 CCA at the first three harmonics of 12 Hz (user interface: row-shaped arrangement). Red bar: target frequency.

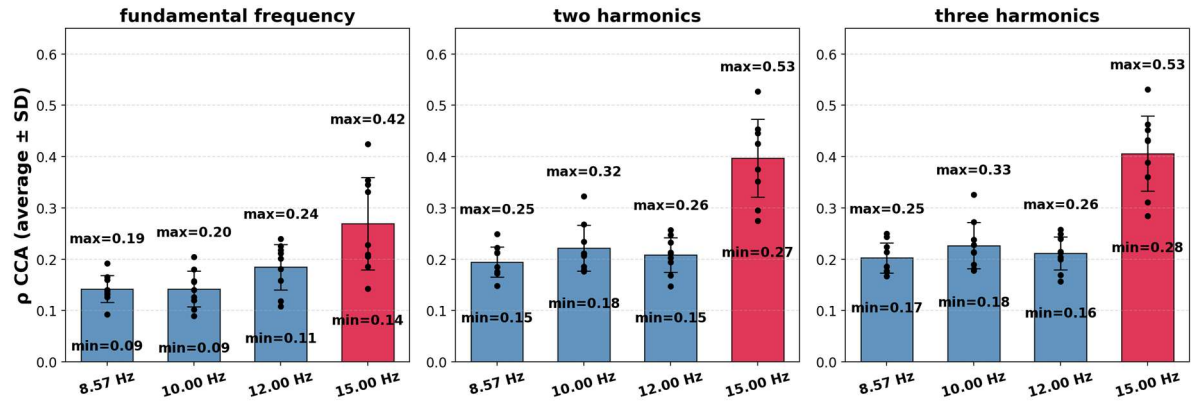


Figure A.2. 9 CCA at the first three harmonics of 15 Hz (user interface: row-shaped arrangement). Red bar: target frequency.

APPENDIX 3 - Use of Artificial Intelligence

Along with the development of this Bachelor Thesis, it has been used as a complementary tool the use of AIs linked to brainstorming, grammar corrections and as a guide for software implementation. For the online BCI repository logo, diagram sketching ideas and generation of images used in Figure 26 from this document and for the PowerPoint presentation, ChatGPT was used.

The provided software was inspired by existing applications from GitHub repositories listed in Appendix 1, and the use of Claude AI was contemplated to help with debugging and brainstorming the combination of different projects with the available OpenBCI hardware equipment.

The development of this document was achieved by the student with the corrections and guidance from the supervisor, as well as the process of experimental trials until an optimal system was achieved. Therefore, the student comprehends and maintains a critical perspective of the benefits and flaws of the final project version and understands the responsibility of developing a Bachelor Thesis using AI as a complementary tool.

Therefore, the whole project was developed by the student, with the use of bibliographic tools such as Zotero, online inspiration sites like GitHub, sketching diagram ideas with Canva, grammar and orthographic reviews by using DeepL and Grammarly, brainstorming with ChatGPT and Claude AI regarding BCIs applications and code debugging with Claude AI; these were utilized to deliver and complete the project for the Early Call placed in September. Also, knowledge from the curricular internship was considered to build up this project.

The initial software repository where different development versions were included is this one: [CarmenEngBio/ssvep-basic](#); but the consideration for the final online assistive SSVEP BCI system is this other one: [CarmenEngBio/mindaaid-ssvep-bci: MindAid Assistive SSVEP BCI. Assistive SSVEP-based Brain-Computer Interface \(BCI\) for real-time selection of basic needs through EEG signals.](#)